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Electrochemistry

JEE Main Chemistry · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	144 questions
Numerical	92 questions
Answer key	at the end

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MCQ (Single Correct Answer) 144 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Open solution

Given at 298 K :

$E^{\ominus}_{Fe^{2+}/Fe} = X$ Volt

$E^{\ominus}_{Fe^{3+}/Fe} = Y$ Volt

The $E^{\ominus}_{Fe^{3+}/Fe^{2+}}$ in Volt at 298 K is given by :

(A) $2X - 3Y$ (B) $3Y - 2X$ (C) $3Y + 2X$ (D) $Y + X$

Q2. JEE Main 2026 (Online) 6th April Evening Shift

Open solution

For a general redox reaction

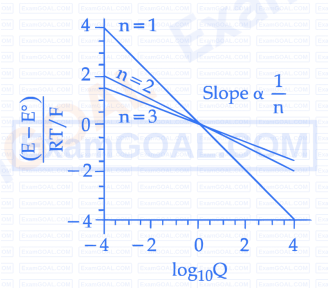
Anode $Red_1 \rightarrow Ox_1^{n_1^+} + n_1e^-$

Cathode $Ox_2 + n_2e^- \rightarrow Red_2^{n_2^-}$

Which of the following statement is incorrect?

(A) The overall reaction can be written as $n_2\,Red_1 + n_1Ox_2 \rightleftharpoons n_2Ox_1^{n_1^+} + n_1\,Red_2^{n_2^-}$

(B) The electrons do not appear in the overall reaction because electrons produced at the anode are consumed at the cathode.



(C) Here n is the number of electrons transferred in redox reaction.

(D) If the reaction is carried out reversibly, the electrical work done is equal to the ratio of charge and potential difference through which charge is moved.

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Q3. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

Consider the following data.

Electrolyte	$\wedge_m^\circ (\mathbf{S\ cm^2\ mol^{-1}})$
$BaCl_2$	x_1
H_2SO_4	x_2
HCl	x_3

$BaSO_4$ is sparingly soluble in water. If the conductivity of the saturated $BaSO_4$ solution is $x\ S\ cm^{-1}$ then the solubility product of $BaSO_4$ can be given as (Here $\wedge_m = \wedge_m^\circ$)

(A) $\frac{10^6 x^2}{\alpha^2(x_1+x_2-2x_3)^2}$

(B) $\frac{x^2}{(x_1+x_2-2x_3)^2}$

(C) $\frac{\alpha^2(x_1+x_2-2x_3)^2}{10^6 x^2}$

(D) $\frac{x^2}{(x_1+x_2+2x_3)^2}$

Q4. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

One half cell in a voltaic cell is constructed by dipping silver rod in $AgNO_3$ solution of unknown concentration, other half cell is Zn rod dipped in 1 molar solution of $ZnSO_4$.

A voltage of 1.60 V is measured at 298 K for this cell. What is the concentration of Ag^+ions used in terms of $\log x$ $(x = [Ag^+])$?

$E^\ominus_{Zn^{2+}/Zn} = -0.76\ V, \quad E^\ominus_{Ag^+/Ag} = +0.80\ V, \frac{2.303RT}{F} = 0.059\ V$

(A) $\frac{2}{3.9}$

(B) $\frac{4}{5.9}$

(C) $\frac{2.9}{2}$

(D) $\frac{5.9}{4}$

Q5. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

An electrochemical cell is constructed using half cells in the direction of spontaneous change:

$Fe(OH)_2(s) + 2e^- \rightarrow Fe(s) + 2OH^-(aq) \quad E^\theta = -0.88\ V$

and $AgBr(s) + e^- \rightarrow Ag(s) + Br^-(aq) \quad E^\theta = +0.07\ V$

Which of the following options is **correct**?

(A) Overall reaction $Fe(s) + 2OH^-(aq) + 2AgBr(s) \rightleftharpoons Fe(OH)_2(s) + 2Ag(s) + 2Br^-(aq)$

(B) $E^\theta_{cell} = -0.95\ V$

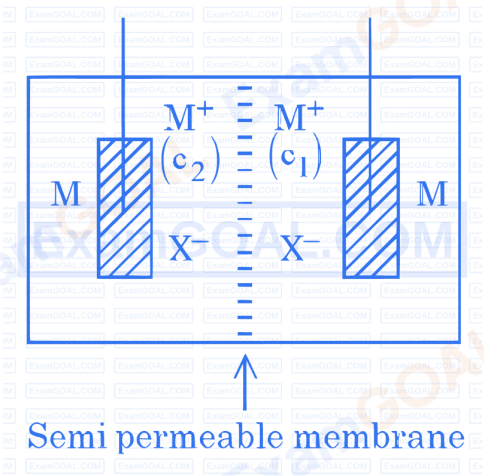
(C) Fe is reduced in the electrochemical cell

(D) E^θ_{cell} is an extensive property

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Q6. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution



Consider the above electrochemical cell where a metal electrode (M) is undergoing redox reaction by forming M^+ $(M \rightarrow M^+ + e^-)$. The cation $M^+$is present in two different concentrations c_1 and c_2 as shown above. Which of the following statement is correct for generating a positive cell potential?

(A) If c_1 is present at anode, then $c_1 > c_2$.

(B) If c_1 is present at cathode, then $c_1 > c_2$.

(C) If c_1 is present at cathode, then $c_1 < c_2$.

(D) If c_1 is present at anode, then $c_1 = c_2$.

Q7. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

In the given electrochemical cell, $Ag(s)|AgCl(s)|FeCl_2(aq), FeCl_3(aq) \mid Pt(s)$ at 298 K , the cell potential (E_{cell}) will increase when :

A. Concentration of Fe^{2+} is increased.

B. Concentration of Fe^{3+} is decreased.

C. Concentration of Fe^{2+} is decreased.

D. Concentration of Fe^{3+} is increased.

E. Concentration of $Cl^-$is increased.

Choose the correct answer from the options given below :

(A) C, D and E Only (B) A and B Only (C) B Only (D) A and E Only

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Q8. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Consider the following reduction processes :

$Al^{3+} + 3e^- \longrightarrow Al(s), E^0 = -1.66\ V$

$Fe^{3+} + e^- \longrightarrow Fe^{2+}, E^0 = +0.77\ V$

$Co^{3+} + e^- \longrightarrow Co^{2+}, E^0 = +1.81\ V$

$Cr^{3+} + 3e^- \longrightarrow Cr(s), E^0 = -0.74\ V$

The tendency to act as reducing agent decreases in the order :

(A) $Al > Fe^{2+} > Cr > Co^{2+}$

(B) $Al > Cr > Co^{2+} > Fe^{2+}$

(C) $Cr > Fe^{2+} > Al > Co^{2+}$

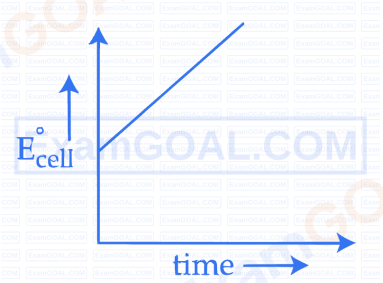
(D) $Al > Cr > Fe^{2+} > Co^{2+}$

Q9. JEE Main 2026 (Online) 21st January Evening Shift

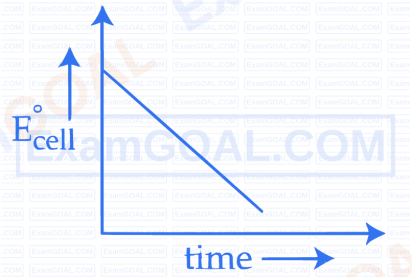
Open solution

For a closed circuit Daniell cell, which of the following plots is the accurate one at a given temperature?

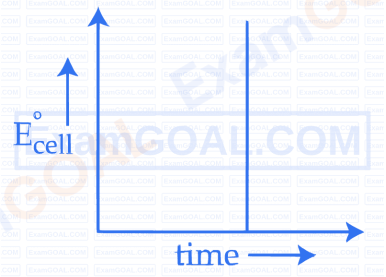
(A)



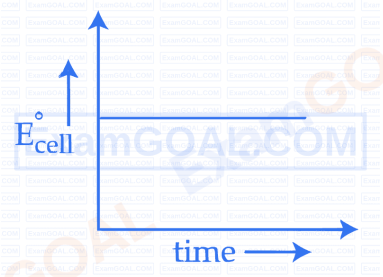
(B)



(C)



(D)



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Q10. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

Given below are two statements :

1 M aqueous solutions of each of $Cu(NO_3)_2$, $AgNO_3$, $Hg_2(NO_3)_2$, $Mg(NO_3)_2$ are electrolysed using inert electrodes. Given: $E^0_{Ag^+/Ag}$ = 0.80 V, $E^0_{Hg_2^{2+}/Hg}$ = 0.79 V, $E^0_{Cu^{2+}/Cu}$ = 0.24 V and $E^0_{Mg^{2+}/Mg}$ = -2.37 V.

Statement (I) : With increasing voltage, the sequence of deposition of metals on the cathode will be Ag, Hg and Cu.

Statement (II) : Magnesium will not be deposited at the cathode instead oxygen gas will be evolved at the cathode.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both **Statement I** and **Statement II** are incorrect

(B) **Statement I** is incorrect but **Statement II** is correct

(C) **Statement I** is correct but **Statement II** is incorrect

(D) Both **Statement I** and **Statement II** are correct

Q11. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

On charging the lead storage battery, the oxidation state of lead changes from x_1 to y_1 at the anode and from x_2 to y_2 at the cathode. The values of x_1, y_1, x_2, y_2 are respectively :

(A) $0, +2, +4, +2$ (B) $+2, 0, 0, +4$ (C) $+4, +2, 0, +2$ (D) $+2, 0, +2, +4$

Q12. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

The standard cell potential $\left(E^{\ominus}_{cell}\right)$ of a fuel cell based on the oxidation of methanol in air that has been used to power television relay station is measured as 1.21 V . The standard half cell reduction potential for O_2 $\left(E^{\circ}_{O_2/H_2O}\right)$ is 1.229 V .

Choose the correct statement :

(A) Reactants are fed at one go to each electrode.

(B) Oxygen is formed at the anode.

(C) The standard half cell reduction potential for the reduction of CO_2 $\left(E^{\circ}_{CO_2/CH_3OH}\right)$ is 19 mV

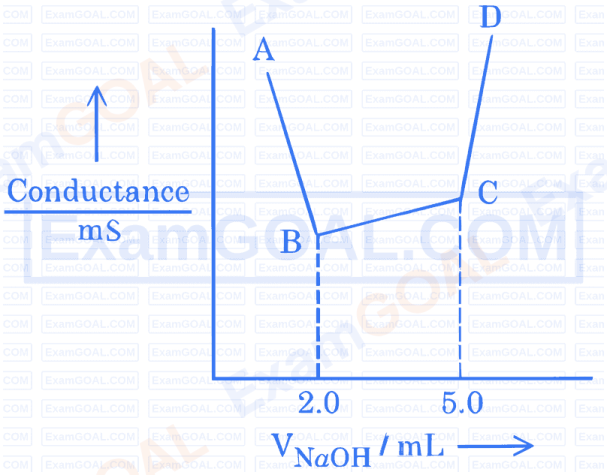
(D) Reduction of methanol takes place at the cathode.

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Q13. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

40 mL of a mixture of CH_3COOH and HCl (aqueous solution) is titrated against 0.1 M NaOH solution conductometrically. Which of the following statement is correct?



(A) The concentration of CH_3COOH in the original mixture is 0.005 M

(B) The concentration of HCl in the original mixture is 0.005 M

(C) CH_3COOH is neutralised first followed by neutralisation of HCl

(D) Point ' C ' indicates the complete neutralisation of HCl

Q14. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

Correct order of limiting molar conductivity for cations in water at 298 K is :

(A) $H^+ > Na^+ > K^+ > Ca^{2+} > Mg^{2+}$

(B) $H^+ > Ca^{2+} > Mg^{2+} > K^+ > Na^+$

(C) $Mg^{2+} > H^+ > Ca^{2+} > K^+ > Na^+$

(D) $H^+ > Na^+ > Ca^{2+} > Mg^{2+} > K^+$

Q15. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

Match List - I with List - II :

List - I (Applications)	List - II (Batteries/Cell)
(A) Transistors	(I) Anode - Zn/Hg; Cathode - HgO + C
(B) Hearing aids	(II) Hydrogen fuel cell
(C) Inverters	(III) Anode - Zn; Cathode - Carbon
(D) Apollo space ship	(IV) Anode - Pb; Cathode - Pb | PbO_2

Choose the correct answer from the options given below :

(A) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(B) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

(C) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(D) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

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Q16. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

O_2 gas will be evolved as a product of electrolysis of :

(A) an aqueous solution of $AgNO_3$ using silver electrodes.

(B) an aqueous solution of $AgNO_3$ using platinum electrodes.

(C) a dilute solution of H_2SO_4 using platinum electrodes.

(D) a high concentration solution of H_2SO_4 using platinum electrodes.

Choose the correct answer from the options given below :

(A) (B) and (C) only **(B)** (B) and (D) only **(C)** (A) and (D) only **(D)** (A) and (C) only

Q17. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

For a Mg | Mg^{2+} (aq) || Ag^{+} (aq) | Ag the correct Nernst Equation is :

(A) $E_{cell} = E^{o}_{cell} - \frac{RT}{2F}\ln\frac{[Ag^{+}]^2}{[Mg^{2+}]}$

(B) $E_{cell} = E^{o}_{cell} - \frac{RT}{2F}\ln\frac{[Mg^{2+}]}{[Ag^{+}]}$

(C) $E_{cell} = E^{o}_{cell} - \frac{RT}{2F}\ln\frac{[Ag^{+}]}{[Mg^{2+}]}$

(D) $E_{cell} = E^{o}_{cell} + \frac{RT}{2F}\ln\frac{[Ag^{+}]^2}{[Mg^{2+}]}$

Q18. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

The molar conductivity of a weak electrolyte when plotted against the square root of its concentration, which of the following is expected to be observed?

(A) Molar conductivity increases sharply with increase in concentration.

(B) Molar conductivity decreases sharply with increase in concentration.

(C) A small increase in molar conductivity is observed at infinite dilution.

(D) A small decrease in molar conductivity is observed at infinite dilution.

Q19. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

The standard reduction potential values of some of the p-block ions are given below. Predict the one with the strongest oxidising capacity.

(A) $E^{o}_{Sn^{4+}/Sn^{2+}} = +1.15$ V

(B) $E^{o}_{Al^{3+}/Al} = -1.66$ V

(C) $E^{o}_{Pb^{4+}/Pb^{2+}} = +1.67$ V

(D) $E^{o}_{Tl^{3+}/Tl} = +1.26$ V

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Q20. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

Based on the data given below :

$E^{\circ}_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33\ V \quad E^{\circ}_{Cl_2/Cl^{(-)}} = 1.36\ V$

$E^{0}_{MnO_4^{-}/Mn^{2+}} = 1.51\ V \quad E^{\circ}_{Cr^{3+}/Cr} = -0.74\ V$

the strongest reducing agent is :

(A) Cl^- (B) MnO_4^- (C) Cr (D) Mn^{2+}

Q21. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

For the given cell

$Fe^{2+}(aq) + Ag^{+}_{(aq)} \rightarrow Fe^{3+}(aq) + Ag_{(s)}$

The standard cell potential of the above reaction is Given:

$Ag^{+} + e^{-} \rightarrow Ag \quad E^{\theta} = xV$

$Fe^{2+} + 2e^{-} \rightarrow Fe \quad E^{\theta} = yV$

$Fe^{3+} + 3e^{-} \rightarrow Fe \quad E^{\theta} = zV$

(A) $x + 2y - 3z$ (B) $x + 2y$ (C) $y - 2x$ (D) $x + y - z$

Q22. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

Standard electrode potentials for a few half cells are mentioned below :

$E^{\circ}_{Cu^{2+}/Cu} = 0.34\ V, E^{\circ}_{Zn^{2+}/Zn} = -0.76\ V$

$E^{\circ}_{Ag^{+}/Ag} = 0.80\ V, E^{\circ}_{Mg^{2+}/Mg} = -2.37\ V$

Which one of the following cells gives the most negative value of ΔG° ?

(A) $Zn\left|Zn^{2+}(1M)\right|\left|Ag^{+}(1M)\right|Ag$

(B) $Ag\left|Ag^{+}(1M)\right|\left|Mg^{2+}(1M)\right|Mg$

(C) $Zn\left|Zn^{2+}(1M)\right|\left|Mg^{2+}(1M)\right|Mg$

(D) $Cu\left|Cu^{2+}(1M)\right|\left|Ag^{+}(1M)\right|Ag$

Q23. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

$FeO_4^{2-} \xrightarrow{+2.0\ V} Fe^{3+} \xrightarrow{0.8\ V} Fe^{2+} \xrightarrow{-0.5\ V} Fe^{0}$

In the above diagram, the standard electrode potentials are given in volts (over the arrow).

The value of $E_{FeO_4^{2-}/Fe^{2+}}$ is :

(A) 1.2 V (B) 2.1 V (C) 1.4 V (D) 1.7 V

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Q24. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

Given below are two statements :

Statement (I) : Corrosion is an electrochemical phenomenon in which pure metal acts as an anode and impure metal as a cathode.

Statement (II) : The rate of corrosion is more in alkaline medium than in acidic medium.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q25. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

Which of the following electrolyte can be used to obtain $H_2S_2O_8$ by the process of electrolysis ?

(A) Concentrated solution of sulphuric acid

(B) Dilute solution of sodium sulphate.

(C) Acidified dilute solution of sodium sulphate.

(D) Dilute solution of sulphuric acid

Q26. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

A solution of aluminium chloride is electrolysed for 30 minutes using a current of 2 A . The amount of the aluminium deposited at the cathode is _____ .

[Given : molar mass of aluminium and chlorine are $27\ g\ mol^{-1}$ and $35.5\ g\ mol^{-1}$ respectively. Faraday constant $= 96500 C\ mol^{-1}$]

(A) 1.660 g (B) 1.007 g (C) 0.336 g (D) 0.441 g

Q27. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

Match List I with List II

	LIST I (Cell)		LIST II (Use/Property/Reaction)
A.	Leclanche cell	I.	Converts energy of combustion into electrical energy
B.	Ni - Cd cell	II.	Does not involve any ion in solution and is used in hearing aids
C.	Fuel cell	III.	Rechargeable
D.	Mercury cell	IV.	Reaction at anode $Zn \rightarrow Zn^{2+} + 2e^-$

Choose the correct answer from the options given below :

(A) A-III, B-I, C-IV, D-II (B) A-I, B-II, C-III, D-IV

(C) A-IV, B-III, C-I, D-II (D) A-II, B-III, C-IV, D-I

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Q28. JEE Main 2024 (Online) 9th April Evening Shift

Which out of the following is a correct equation to show change in molar conductivity with respect to concentration for a weak electrolyte, if the symbols carry their usual meaning :

Open solution

(A) $\Lambda_m - \Lambda_m^\circ + AC^{\frac{1}{2}} = 0$

(B) $\Lambda_m^2 C + K_a \Lambda_m^{o^2} - K_a \Lambda_m \Lambda_m^\circ = 0$

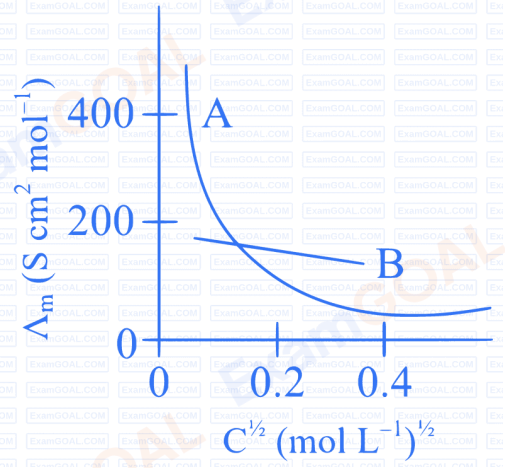
(C) $\Lambda_m - \Lambda_m^\circ - AC^{\frac{1}{2}} = 0$

(D) $\Lambda_m^2 C - K_a \Lambda_m^{\circ 2} + K_a \Lambda_m \Lambda_m^\circ = 0$

Q29. JEE Main 2024 (Online) 9th April Morning Shift

The molar conductivity for electrolytes A and B are plotted against $C^{1/2}$ as shown below. Electrolytes A and B respectively are:

Open solution



(A) A - weak electrolyte B - weak electrolyte

(B) A - weak electrolyte B - strong electrolyte

(C) A - strong electrolyte B - strong electrolyte

(D) A - strong electrolyte B - weak electrolyte

Q30. JEE Main 2024 (Online) 8th April Evening Shift

The emf of cell $Tl \left| \underset{(0.001M)}{Tl^+} \right| \underset{(0.01M)}{Cu^{2+}} \mid Cu$ is $0.83\ V$ at $298\ K$. It could be increased by :

Open solution

(A) increasing concentration of Tl^+ ions

(B) increasing concentration of Cu^{2+} ions

(C) increasing concentration of both Tl^+ and Cu^{2+} ions

(D) decreasing concentration of both Tl^+ and Cu^{2+} ions

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Q31. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

The reaction;

$\frac{1}{2}H_{2(g)} + AgCl_{(s)} \rightarrow H^{+}_{(aq)} + Cl^{-}_{(aq)} + Ag_{(s)}$

occurs in which of the following galvanic cell :

(A) $Pt\left|H_{2(g)}\right|HCl_{(soln.)}\left|AgNO_{3(aq)}\right|Ag$

(B) $Ag\left|AgCl_{(s)}\right|KCl_{(soln.)}\left|AgNO_{3\,(aq.)}\right|Ag$

(C) $Pt\left|H_{2(g)}\right|KCl_{(soln.)}\left|AgCl_{(s)}\right|Ag$

(D) $Pt\left|H_{2(g)}\right|HCl_{(soln.)}\left|AgCl_{(s)}\right|Ag$

Q32. JEE Main 2024 (Online) 6th April Evening Shift

Open solution

How can an electrochemical cell be converted into an electrolytic cell ?

(A) Applying an external opposite potential greater than E^{0}_{cell}

(B) Exchanging the electrodes at anode and cathode.

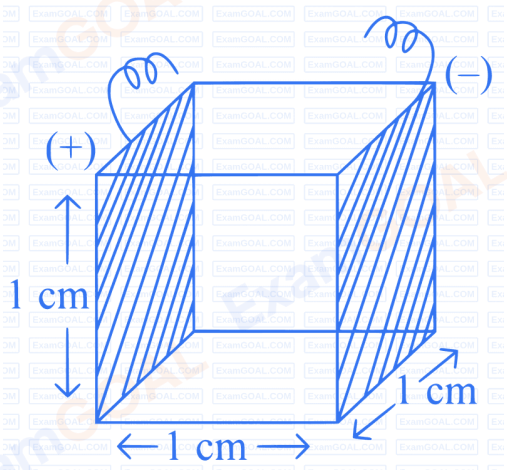
(C) Applying an external opposite potential lower than E^{0}_{cell}

(D) Reversing the flow of ions in salt bridge.

Q33. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

A conductivity cell with two electrodes (dark side) are half filled with infinitely dilute aqueous solution of a weak electrolyte. If volume is doubled by adding more water at constant temperature, the molar conductivity of the cell will -



(A) depend upon type of electrolyte

(B) increase sharply

(C) decrease sharply

(D) remain same or can not be measured accurately

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Q34. JEE Main 2024 (Online) 5th April Evening Shift

The quantity of silver deposited when one coulomb charge is passed through $AgNO_3$ solution :

(A) 0.1 g atom of silver

(B) 1 chemical equivalent of silver

(C) 1 g of silver

(D) 1 electrochemical equivalent of silver

Open solution

Q35. JEE Main 2024 (Online) 5th April Evening Shift

For the electrochemical cell

$M\left|M^{2+}\right|\left|X\right|X^{2-}$

If $E^0_{(M^{2+}/M)} = 0.46\ V$ and $E^0_{(X/X^{2-})} = 0.34\ V$.

Which of the following is correct?

(A) $E_{cell} = 0.80\ V$

(B) $M + X \rightarrow M^{2+} + X^{2-}$ is a spontaneous reaction

(C) $E_{cell} = -0.80\ V$

(D) $M^{2+} + X^{2-} \rightarrow M + X$ is a spontaneous reaction

Open solution

Q36. JEE Main 2024 (Online) 5th April Morning Shift

Molar ionic conductivities of divalent cation and anion are $57\ S\ cm^2\ mol^{-1}$ and $73\ S\ cm^2\ mol^{-1}$ respectively. The molar conductivity of solution of an electrolyte with the above cation and anion will be:

(A) $187\ S\ cm^2\ mol^{-1}$

(B) $260\ S\ cm^2\ mol^{-1}$

(C) $65\ S\ cm^2\ mol^{-1}$

(D) $130\ S\ cm^2\ mol^{-1}$

Open solution

Q37. JEE Main 2024 (Online) 5th April Morning Shift

The reaction at cathode in the cells commonly used in clocks involves.

(A) oxidation of Mn from +2 to +7

(B) reduction of Mn from +4 to +3

(C) oxidation of Mn from +3 to +4

(D) reduction of Mn from +7 to +2

Open solution

Q38. JEE Main 2024 (Online) 4th April Evening Shift

Fuel cell, using hydrogen and oxygen as fuels,

A. has been used in spaceship

B. has an efficiency of 40% to produce electricity

C. uses aluminum as catalysts

D. is eco-friendly

E. is actually a type of Galvanic cell only

Choose the correct answer from the options given below:

(A) A, D, E only

(B) A, B, C only

(C) A, B, D only

(D) A, B, D, E only

Open solution

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Q39. JEE Main 2024 (Online) 4th April Evening Shift

For a strong electrolyte, a plot of molar conductivity against (concentration) $^{1/2}$ is a straight line, with a negative slope, the correct unit for the slope is

(A) $\mathrm{Scm^2\ mol^{-3/2}\ L}$

(B) $\mathrm{Scm^2\ mol^{-3/2}\ L^{-1/2}}$

(C) $\mathrm{Scm^2\ mol^{-1}\ L^{1/2}}$

(D) $\mathrm{Scm^2\ mol^{-3/2}\ L^{1/2}}$

Open solution

Q40. JEE Main 2024 (Online) 4th April Morning Shift

One of the commonly used electrode is calomel electrode. Under which of the following categories, calomel electrode comes?

(A) Metal ion - Metal electrodes

(B) Oxidation - Reduction electrodes

(C) Metal - Insoluble Salt - Anion electrodes

(D) Gas - Ion electrodes

Open solution

Q41. JEE Main 2024 (Online) 4th April Morning Shift

What pressure (bar) of $\mathrm{H_2}$ would be required to make emf of hydrogen electrode zero in pure water at $25^{\circ}\mathrm{C}$?

(A) 0.5 (B) 10^{-14} (C) 1 (D) 10^{-7}

Open solution

Q42. JEE Main 2024 (Online) 31st January Morning Shift

Identify the factor from the following that does not affect electrolytic conductance of a solution.

(A) The nature of solvent used.

(B) The nature of the electrolyte added.

(C) The nature of the electrode used.

(D) Concentration of the electrolyte.

Open solution

Q43. JEE Main 2024 (Online) 30th January Evening Shift

Reduction potential of ions are given below:

$\mathrm{ClO_4^-}$	$\mathrm{IO_4^-}$	$\mathrm{BrO_4^-}$
$\mathrm{E^{\circ} = 1.19\ V}$	$\mathrm{E^{\circ} = 1.65\ V}$	$\mathrm{E^{\circ} = 1.74\ V}$

The correct order of their oxidising power is :

(A) $\mathrm{IO_4^- > BrO_4^- > ClO_4^-}$

(B) $\mathrm{BrO_4^- > ClO_4^- > IO_4^-}$

(C) $\mathrm{ClO_4^- > IO_4^- > BrO_4^-}$

(D) $\mathrm{BrO_4^- > IO_4^- > ClO_4^-}$

Open solution

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Q44. JEE Main 2024 (Online) 27th January Evening Shift

Which of the following statements is not correct about rusting of iron?

(A) Rusting of iron is envisaged as setting up of electrochemical cell on the surface of iron object.

(B) Dissolved acidic oxides SO_2, NO_2 in water act as catalyst in the process of rusting.

(C) Coating of iron surface by tin prevents rusting, even if the tin coating is peeling off.

(D) When pH lies above 9 or 10, rusting of iron does not take place.

Open solution

Q45. JEE Main 2023 (Online) 12th April Morning Shift

For lead storage battery pick the correct statements

A. During charging of battery, $PbSO_4$ on anode is converted into PbO_2

B. During charging of battery, $PbSO_4$ on cathode is converted into PbO_2

C. Lead storage battery consists of grid of lead packed with PbO_2 as anode

D. Lead storage battery has $\sim 38\%$ solution of sulphuric acid as an electrolyte

Choose the correct answer from the options given below:

(A) A, B, D only (B) B, C only (C) B, C, D only (D) B, D only

Open solution

Q46. JEE Main 2023 (Online) 8th April Morning Shift

The reaction

$\frac{1}{2}H_2(g) + AgCl(s) \rightleftharpoons H^+(aq) + Cl^-(aq) + Ag(s)$

occurs in which of the given galvanic cell.

(A) $Ag|AgCl(s)|KCl\,(sol^n)\,|AgNO_3|Ag$

(B) $Pt\,|H_2(g)|HCl\,(sol^n)|AgCl(s)|Ag$

(C) $Pt\,|H_2(g)|HCl\,(sol^n)\,|AgNO_3\,(sol^n)|Ag$

(D) $Pt\,|H_2(g)|KCl\,(sol^n)|AgCl(s)|Ag$

Open solution

Q47. JEE Main 2023 (Online) 6th April Morning Shift

The standard electrode potential of M^+/M in aqueous solution does not depend on

(A) Ionisation of a gaseous metal atom (B) Sublimation of a solid metal

(C) Ionisation of a solid metal atom (D) Hydration of a gaseous metal ion

Open solution

Q48. JEE Main 2023 (Online) 31st January Morning Shift

Which one of the following statements is correct for electrolysis of brine solution?

(A) O_2 is formed at cathode (B) H_2 is formed at anode

(C) Cl_2 is formed at cathode (D) OH^- is formed at cathode

Open solution

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Q49. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

The standard electrode potential (M^{3+}/M^{2+}) for V, Cr, Mn & Co are -0.26 V, -0.41 V, + 1.57 V and + 1.97 V, respectively. The metal ions which can liberate H_2 from a dilute acid are :

(A) Mn^{2+} and Co^{2+}

(B) V^{2+} and Mn^{2+}

(C) V^{2+} and Cr^{2+}

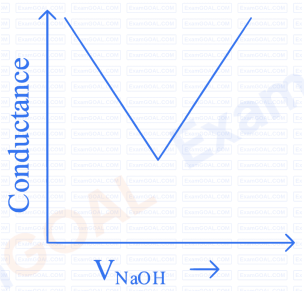
(D) Cr^{2+} and Co^{2+}

Q50. JEE Main 2023 (Online) 24th January Evening Shift

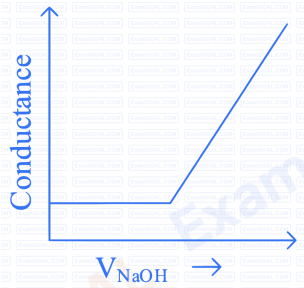
Open solution

Choose the correct representation of conductometric titration of benzoic acid vs sodium hydroxide.

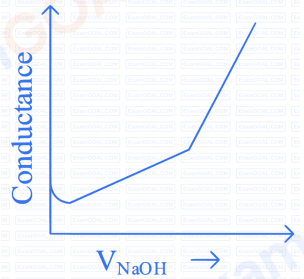
(A)



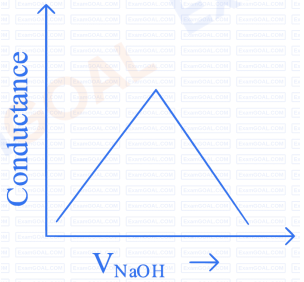
(B)



(C)



(D)



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Q51. JEE Main 2022 (Online) 30th July Evening Shift

Open solution

The E° (volt) values are : Al^{3+}/Al, -1.66, Sc^{3+}/Sc, -2.08, Fe^{3+}/Fe^{2+}, $+0.77$, Hg_2^{2+}/Hg, $+0.79$. Arrangement of cations, Al^{3+}, Sc^{3+}, Fe^{3+} and Hg_2^{2+} in decreasing order of oxidizing strength is

(A) $Al^{3+} > Sc^{3+} > Fe^{3+} > Hg_2^{2+}$

(B) $Hg_2^{2+} > Fe^{3+} > Sc^{3+} > Al^{3+}$

(C) $Hg_2^{2+} > Fe^{3+} > Al^{3+} > Sc^{3+}$

(D) $Sc^{3+} > Al^{3+} > Fe^{3+} > Hg_2^{2+}$

Q52. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

Match List - I with List - II.

	List - I		List - II
(A)	$Cd(s) + 2Ni(OH)_3(s) \rightarrow CdO(s) + 2Ni(OH)_2(s) + H_2O(l)$	(I)	Primary battery
(B)	$Zn(Hg) + HgO(s) \rightarrow ZnO(s) + Hg(l)$	(II)	Discharging of secondary battery
(C)	$2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)$	(III)	Fuel cell
(D)	$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$	(IV)	Charging of secondary battery

Choose the correct answer from the options given below:

(A) $(A) - (I), (B) - (II), (C) - (III), (D) - (IV)$

(B) $(A) - (IV), (B) - (I), (C) - (II), (D) - (III)$

(C) $(A) - (II), (B) - (I), (C) - (IV), (D) - (III)$

(D) $(A) - (II), (B) - (I), (C) - (III), (D) - (IV)$

Q53. JEE Main 2022 (Online) 27th July Evening Shift

Open solution

Given below are two statements :

Statement I : For KI, molar conductivity increases steeply with dilution

Statement II : For carbonic acid, molar conductivity increases slowly with dilution

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q54. JEE Main 2022 (Online) 25th July Evening Shift

Open solution

The molar conductivity of a conductivity cell filled with 10 moles of 20 mL NaCl solution is Λ_{m1} and that of 20 moles another identical cell having 80 mL NaCl solution is Λ_{m2}. The conductivities exhibited by these two cells are same. The relationship between Λ_{m2} and Λ_{m1} is

(A) $\Lambda_{m2} = 2\Lambda_{m1}$

(B) $\Lambda_{m2} = \Lambda_{m1} / 2$

(C) $\Lambda_{m2} = \Lambda_{m1}$

(D) $\Lambda_{m2} = 4\Lambda_{m1}$

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Q55. JEE Main 2022 (Online) 30th June Morning Shift

In which of the following half cells, electrochemical reaction is pH dependent?

(A) $Pt\,|\,Fe^{3+},\,Fe^{2+}$ (B) $MnO_4^-\,|Mn^{2+}$ (C) $Ag\,|\,AgCl\,|Cl^{-1}$ (D) $\frac{1}{2}F_2\,|F^-$

Open solution

Q56. JEE Main 2022 (Online) 26th June Morning Shift

The $\left(\frac{\partial E}{\partial T}\right)_P$ of different types of half cells are as follows:

A	B	C	D
1×10^{-4}	2×10^{-4}	0.1×10^{-4}	0.2×10^{-4}

(Where E is the electromotive force)

Which of the above half cells would be preferred to be used as reference electrode?

(A) A (B) B (C) C (D) D

Open solution

Q57. JEE Main 2022 (Online) 25th June Evening Shift

The correct order of reduction potentials of the following pairs is

A. Cl_2/Cl^-

B. I_2/I^-

C. Ag^+/Ag

D. Na^+/Na

E. Li^+/Li

Choose the correct answer from the options given below.

(A) A > C > B > D > E (B) A > B > C > D > E

(C) A > C > B > E > D (D) A > B > C > E > D

Open solution

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Q58. JEE Main 2021 (Online) 31st August Evening Shift

Match List - I with List - II

Open solution

	List - I (Parameter)		List - II (Unit)
(a)	Cell constant	(i)	$S\,cm^2mol^{-1}$
(b)	Molar conductivity	(ii)	Dimensionless
(c)	Conductivity	(iii)	m^{-1}
(d)	Degree of dissociation of electrolyte	(iv)	$\Omega^{-1}m^{-1}$

Choose the most appropriate answer from the options given below :

(A) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(B) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(C) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

(D) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

Q59. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

Given below are two statements :

Statement I : The limiting molar conductivity of KCl (strong electrolyte) is higher compared to that of CH_3COOH (weak electrolyte).

Statement II : Molar conductivity decreases with decrease in concentration of electrolyte.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is true but Statement II is false.

(B) Statement I is false but Statement II is true.

(C) Both Statement I and Statement II are true.

(D) Both Statement I and Statement II are false.

Q60. JEE Main 2021 (Online) 26th February Morning Shift

Open solution

Compound A used as a strong oxidizing agent is amphoteric in nature. It is the part of lead storage batteries. Compound A is :

(A) PbO **(B)** PbO_2 **(C)** $PbSO_4$ **(D)** Pb_3O_4

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Q61. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

For the given cell :

$Cu(s) \mid Cu^{2+}(C_1M) \parallel Cu^{2+}(C_2M) \mid Cu(s)$

change in Gibbs energy (ΔG) is negative, if :

(A) $C_2 = \sqrt{2}C_1$

(B) $C_2 = \frac{C_1}{\sqrt{2}}$

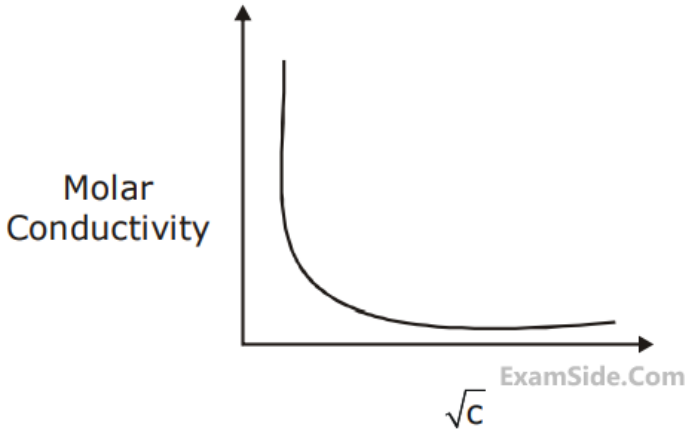
(C) $C_1 = 2C_2$

(D) $C_1 = C_2$

Q62. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

The variation of molar conductivity with concentration of an electrolyte (X) in aqueous solution is shown in the given figure.



The electrolyte X is :

(A) HCl

(B) CH_3COOH

(C) NaCl

(D) KNO_3

Q63. JEE Main 2020 (Online) 4th September Evening Slot

Open solution

250 mL of a waste solution obtained from the workshop of a goldsmith contains 0.1 M $AgNO_3$ and 0.1 M AuCl. The solution was electrolyzed at 2V by passing a current of 1A for 15 minutes. The metal/metals electrodeposited will be

[$E^0_{Ag^+/Ag}$ = 0.80 V, $E^0_{Au^+/Au}$ = 1.69 V]

(A) Silver and gold in equal mass proportion

(B) Silver and gold in proportion to their atomic weights

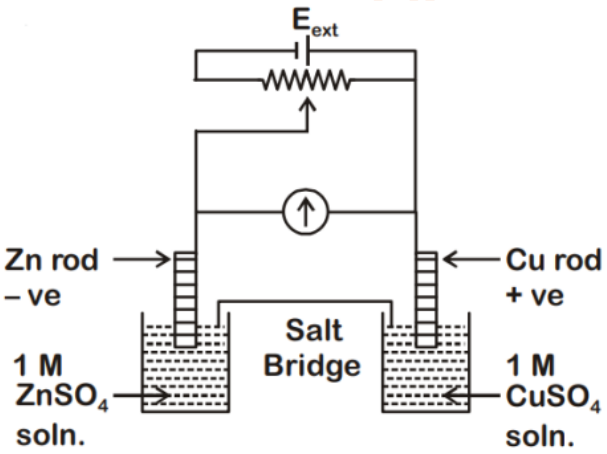
(C) Only gold

(D) Only silver

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Q64. JEE Main 2020 (Online) 4th September Morning Slot

Open solution



$E^0_{Cu^{2+}|Cu}$ = +0.34 V

$E^0_{Zn^{2+}|Zn}$ = -0.76 V

Identify the incorrect statement from the option below for the above cell :

(A) If E_{ext} < 1.1 V, Zn dissolves at anode and Cu deposits at cathode

(B) If E_{ext} = 1.1 V, no flow of e^- or current occurs

(C) If E_{ext} > 1.1 V, e^- flows from Cu to Zn

(D) If E_{ext} > 1.1 V, Zn dissolves at Zn electrode and Cu deposits at Cu electrode

Q65. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

Let C_{NaCl} and C_{BaSO_4} be the conductances (in S) measured for saturated aqueous solutions of NaCl and $BaSO_4$, respectively, at a temperature T. Which of the following is false?

(A) Ionic mobilities of ions from both salts increase with T.

(B) $C_{NaCl}(T_2) > C_{NaCl}(T_1)$ for $T_2 > T_1$

(C) $C_{BaSO_4}(T_2) > C_{BaSO_4}(T_1)$ for $T_2 > T_1$

(D) $C_{NaCl} >> C_{BaSO_4}$ at a given T

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Q66. JEE Main 2020 (Online) 7th January Evening Slot

Open solution

The equation that is incorrect is :

(A) $(\Lambda_m^0)_{KCl} - (\Lambda_m^0)_{NaCl} = (\Lambda_m^0)_{KBr} - (\Lambda_m^0)_{NaBr}$

(B) $(\Lambda_m^0)_{NaBr} - (\Lambda_m^0)_{NaI} = (\Lambda_m^0)_{KBr} - (\Lambda_m^0)_{NaBr}$

(C) $(\Lambda_m^0)_{NaBr} - (\Lambda_m^0)_{NaCl} = (\Lambda_m^0)_{KBr} - (\Lambda_m^0)_{KCl}$

(D) $(\Lambda_m^0)_{H_2O} = (\Lambda_m^0)_{HCl} + (\Lambda_m^0)_{NaOH} - (\Lambda_m^0)_{NaCl}$

Q67. JEE Main 2020 (Online) 7th January Morning Slot

Open solution

Given that the standard potentials (E^o) of Cu^{2+}/Cu and Cu^+/Cu are 0.34 V and 0.522 V respectively, the E^o of Cu^{2+}/Cu^+ :

(A) - 0.182 V (B) - 0.158 V (C) 0.182 V (D) +0.158 V

Q68. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

Given

$Co^{3+} + e^- \rightarrow Co^{2+}$; E^o = + 1.81 V

$Pb^{4+} + 2e^- \rightarrow Pb^{2+}$; E^o = + 1.67 V

$Ce^{4+} + e^- \rightarrow Ce^{3+}$; E^o = + 1.61 V

$Bi^{3+} + 3e^- \rightarrow Bi$; E^o = + 0.20 V

Oxidizing power of the species will increase in the order :

(A) $Co^{3+} < Ce^{4+} < Bi^{3+} < Pb^{4+}$

(B) $Co^{3+} < Pb^{4+} < Ce^{4+} < Bi^{3+}$

(C) $Ce^{4+} < Pb^{4+} < Bi^{3+} < Co^{3+}$

(D) $Bi^{3+} < Ce^{4+} < Pb^{4+} < Co^{3+}$

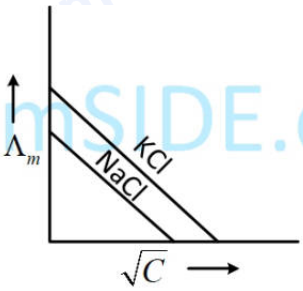
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Q69. JEE Main 2019 (Online) 10th April Evening Slot

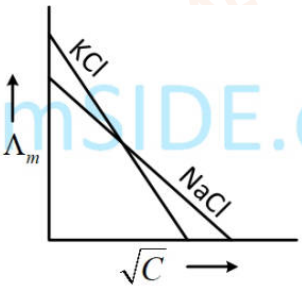
Open solution

Which one of the following graphs between molar conductivity (Λ_m) versus $\sqrt{C}$ is correct ?

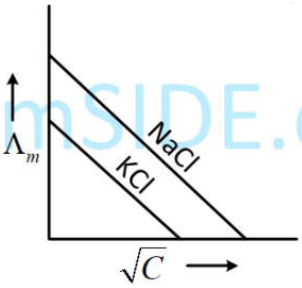
(A)



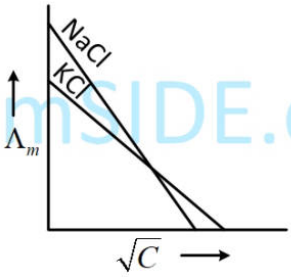
(B)



(C)



(D)



Q70. JEE Main 2019 (Online) 10th April Morning Slot

Open solution

Consider the statements S1 and S2

S1 : Conductivity always increases with decrease in the concentration of electrolyte.

S2 : Molar conductivity always increases with decrease in the concentration of electrolyte.

The correct option among the following is :

(A) Both S1 and S2 are wrong

(B) S1 is correct and S2 is wrong

(C) Both S1 and S2 are correct

(D) S1 is wrong and S2 is correct

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Q71. JEE Main 2019 (Online) 9th April Evening Slot

A solution of $Ni(NO_3)_2$ is electrolysed between platinum electrodes using 0.1 Faraday electricity. How many moles of Ni will be deposited at the cathode?

(A) 0.10 (B) 0.15 (C) 0.20 (D) 0.05

Open solution

Q72. JEE Main 2019 (Online) 9th April Morning Slot

The standard Gibbs energy for the given cell reaction in kJ mol^{-1} at 298 K is :

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$,

$E° = 2$ V at 298 K

(Faraday's constant, F = 96000 C mol^{-1})

(A) 384 (B) −192 (C) −384 (D) 192

Open solution

Q73. JEE Main 2019 (Online) 8th April Evening Slot

Calculate the standard cell potential in (V) of the cell in which following reaction takes place :

$Fe^{2+}(aq) + Ag^{+}(aq) \rightarrow Fe^{3+}(aq) + Ag(s)$

Given that

$E^o_{Ag^+/Ag} = xV$

$E^o_{Fe^{2+}/Fe} = yV$

$E^o_{Fe^{3+}/Fe} = zV$

(A) x + 2y - 3z (B) x - z (C) x - y (D) x + y - z

Open solution

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Q74. JEE Main 2019 (Online) 8th April Morning Slot

Open solution

Given that $E^{\Theta}_{O_2/H_2O} = 1.23\,V$;

$E^{\Theta}_{S_2O_8^{2-}/SO_4^{2-}} = 2.05\,V$

$E^{\Theta}_{Br_2/Br^-} = 1.09\,V$

$E^{\Theta}_{Au^{3+}/Au} = 1.4\,V$

The strongest oxidizing agent is :

(A) O_2 (B) Au^{3+} (C) Br_2 (D) $S_2O_8^{2-}$

Q75. JEE Main 2019 (Online) 12th January Evening Slot

Open solution

$\wedge_m^\circ$ for NaCl, HCl and NaA are 126.4, 425.9 and 100.5 S cm^2 mol^{-1}, respectively. If the conductivity of 0.001 M HA is-

5×10^{-5} S cm^{-1}, degree of dissociation of HA is -

(A) 0.50 (B) 0.125 (C) 0.25 (D) 0.75

Q76. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

The standard electrode potential E^o and its temperature coefficient $\left(\frac{dE^o}{dT}\right)$ for a cell are 2V and -5×10^{-4} VK^{-1} at 300 K respectively.
The cell reaction is
$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$

The standard reaction enthalpy (Δ_rH^o) at 300 K in kJ mol^{-1} is, [Use R = 8 JK^{-1} mol^{-1} and F = 96,000C mol^{-1}]

(A) -412.8 (B) -384.0 (C) 192.0 (D) 206.4

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Q77. JEE Main 2019 (Online) 11th January Evening Slot

Open solution

Given the equilibrium constant:

K_C of the reaction :

$Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$ is

10×10^{15}, calculate the E^0_{cell} of this reaction at 298 K

[2.303 $\frac{RT}{F}$ at 298 K = 0.059V]

(A) 0.4736 mV **(B)** 0.04736 V **(C)** 0.4736 V **(D)** 0.04736 mV

Q78. JEE Main 2019 (Online) 11th January Morning Slot

Open solution

For the cell Zn(s) $|Zn^{2+}(aq)|$ $|M^{x+}(aq)|$ M(s), different half cells and their standard electrode potentials are given below :

$M^{x+}(aq)/M(s)$	$Au^{3+}(aq)/Au(s)$	$Ag^+(aq)/Ag(s)$	$Fe^{3+}(aq)/Fe^{2+}(aq)$	$Fe^{2+}(aq)/Fe(s)$
$E^0_{M^{x+}/M}$/(V)	1.40	0.80	0.77	−0.44

If $E^0_{zn^{2+}/zn}$ = − 0.76 V, which cathode will give maximum value of E^o_{cell} per electron transferred?

(A) Ag^+/Ag **(B)** Fe^{3+}/Fe^{2+} **(C)** Au^{3+}/Au **(D)** Fe^{2+}/Fe

Q79. JEE Main 2019 (Online) 10th January Evening Slot

Open solution

In the cell

$Pt|(s)|H_2(g, 1\ bar)|HCl\,(aq)|AgCl|(s)|Ag(s)|Pt(s)$

the cell potential is 0.92 V when a 10^{-6} molal HCl solution is used. The standard electrode potential of (AgCl/Ag/Cl^-) electrode is :
{Given, $\frac{2.303RT}{F} = 0.06V$ at 298}

(A) 0.94 V **(B)** 0.40 V **(C)** 0.76 V **(D)** 0.20 V

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Q80. JEE Main 2019 (Online) 10th January Morning Slot

Open solution

Consider the following reduction processes :

$Zn^{2+} + 2e^- \rightarrow Zn(s)$; $E^o = -0.76$ V

$Ca^{2+} + 2e^- \rightarrow Ca(s)$; $E^o = -2.87$ V

$Mg^{2+} + 2e^- \rightarrow Mg(s)$; $E^o = -2.36$ V

$Ni^2 + 2e^- \rightarrow Ni(s)$; $E^o = -0.25$

The reducing power of the metals increases in the order :

(A) Ca < Mg < Zn < Ni

(B) Ni < Zn < Mg < Ca

(C) Zn < Mg < Ni < Ca

(D) Ca < Zn < Mg < Ni

Q81. JEE Main 2019 (Online) 9th January Evening Slot

Open solution

If the standard electrode potential for a cell is 2 V at 300 K, the equilibrium constant (K) for the reaction

$Zn(s) + Cu^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Cu(s)$

at 300 K is approximately,

($R = 8\ JK^{-1}mol^{-1}$, $F = 96000\ C\ mol^{-1}$)

(A) e^{-80}

(B) e^{-160}

(C) e^{320}

(D) e^{160}

Q82. JEE Main 2019 (Online) 9th January Morning Slot

Open solution

The anodic half-cell of lead-acid battery is recharged using electricity of 0.05 Faraday. The amount of $PbSO_4$ electrolyzed in g during the process is : (Molar mass of $PbSO_4$ = 303 g mol^{-1})

(A) 22.8

(B) 15.2

(C) 7.6

(D) 11.4

Q83. JEE Main 2018 (Online) 16th April Morning Slot

Open solution

When 9.65 ampere current was passed for 1.0 hour into nitrobenzene in acidic medium, the amount of p-aminophenol produced is :

(A) 9.81 g

(B) 10.9 g

(C) 98.1 g

(D) 109.0 g

Q84. JEE Main 2018 (Offline)

Open solution

How long (approximate) should water be electrolysed by passing through 100 amperes current so that the oxygen released can completely burn 27.66 g of diborane?
(Atomic weight of B = 10.8 u)

(A) 1.6 hours

(B) 6.4 hours

(C) 0.8 hours

(D) 3.2 hours

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Q85. JEE Main 2018 (Online) 15th April Morning Slot

When an electric currents passed through acidified water, 112 mL of hydrogen gas at N.T.P. was collected at the cathode in 965 seconds. The current passed, in ampere, is :

(A) 1.0 (B) 0.5 (C) 0.1 (D) 2.0

Open solution

Q86. JEE Main 2017 (Online) 9th April Morning Slot

To find the standard potential of M^{3+}/M electrode, the following cell is constituted : $Pt/M/M^{3+}(0.001\ mol\ L^{-1})/Ag^{+}(0.01\ mol\ L^{-1})/Ag$

The emf of the cell is found to be 0.421 volt at 298 K. The standard potential of half reaction $M^{3+} + 3e^{-} \rightarrow M$ at 298 K will be :

(Given $E^{o}_{Ag^{+}/Ag}$ at 298 K = 0.80 Volt)

(A) 0.38 Volt (B) 0.32 Volt (C) 1.28 Volt (D) 0.66 Volt

Open solution

Q87. JEE Main 2017 (Online) 8th April Morning Slot

What is the standard reduction potential (E^{o}) for $Fe^{3+} \rightarrow Fe$?
Given that :
$Fe^{2+} + 2e^{-} \rightarrow Fe$; $E^{o}_{Fe^{2+}/Fe}$ = −0.47 V
$Fe^{3+} + e^{-} \rightarrow Fe^{2+}$; $E^{o}_{Fe^{3+}/Fe^{2+}}$ = +0.77 V

(A) − 0.057 V (B) + 0.057 V (C) + 0.30 V (D) − 0.30 V

Open solution

Q88. JEE Main 2017 (Online) 8th April Morning Slot

Consider the following standard electrode potentials (E^{o} in volts) in aqueous solution :

Element	M^{3+}/M	M^{+}/M
Al	-1.66	+ 0.55
Tl	+1.26	- 0.34

Based on these data, which of the following statements is **correct** ?

(A) Tl^{+} is more stable than Al^{3+}
(B) Al^{+} is more stable than Al^{3+}
(C) Tl^{+} is more stable than Al^{+}
(D) Tl^{3+} is more stable than Al^{3+}

Open solution

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Q89. JEE Main 2017 (Offline)

Open solution

Given

$E^o_{Cl_2/Cl^-}$ = 1.36 V, $E^o_{Cr^{3+}/Cr}$ = - 0.74 V

$E^o_{Cr_2O_7^{2-}/Cr^{3+}}$ = 1.33 V, $E^o_{MnO_4^-/Mn^{2+}}$ = 1.51 V

Among the following, the strongest reducing agent is :

(A) Mn^{2+} (B) Cr^{3+} (C) Cl^- (D) Cr

Q90. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

Oxidation of succinate ion produces ethylene and carbon dioxide gases. On passing 0.2 Faraday electricity through an aqueous solution of potassium succinate, the total volume of gases (at both cathode and anode) at STP (1 atm and 273 K) is :

(A) 2.24 L (B) 4.48 L (C) 6.72 L (D) 8.96 L

Q91. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

Identify the correct statement :

(A) Iron corrodes in oxygen-free water.

(B) Iron corrodes more rapidly in salt water because its electrochemical potential is higher.

(C) Corrosion of iron can be minimized by forming a contact with another metal with a higher reduction potential.

(D) Corrosion of iron can be minimized by forming an impermeable barrier at its surface.

Q92. JEE Main 2016 (Online) 9th April Morning Slot

Open solution

What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M $ZnSO_4$?

(A) The copper metal will dissolve and zinc metal will be deposited.

(B) The copper metal will dissolve with evolution of hydrogen gas.

(C) The copper metal will dissolve with evolution of oxygen gas.

(D) No reaction will occur.

Q93. JEE Main 2016 (Offline)

Open solution

Galvanization is applying a coating of :

(A) Cr (B) Cu (C) Zn (D) Pb

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Q94. JEE Main 2015 (Online) 11th April Morning Slot

At 298 K, the standard reduction potentials are 1.51 V for $MnO_4^- \,|Mn^{2+}$, 1.36 V for $Cl_2|Cl^-$, 1.07 V for $Br_2|Br^-$, and 0.54 V for $I_2|I^-$. At pH = 3, permanganate is expected to oxidize :
$\left(\frac{2.303RT}{F} = 0.059\,V\right)$

(A) Cl^-, Br^- and I^-
(B) Cl^- and Br^-
(C) Br^- and I^-
(D) I^- only

Open solution

Q95. JEE Main 2015 (Online) 10th April Morning Slot

A variable, opposite external potential (E_{ext}) is applied to the cell

$Zn\,|Zn^{2+}\,(1\,M)|\,|Cu^{2+}\,(1\,M)|\,Cu$, of potential 1.1 V. When E_{ext} < 1.1 V and E_{ext} > 1.1 V, respectively electrons flow from :

(A) anode to cathode and cathode to anode
(B) cathode to anode and anode to cathode
(C) cathode to anode in both cases
(D) anode to cathode in both cases

Open solution

Q96. JEE Main 2015 (Offline)

Two Faraday of electricity is passed through a solution of $CuSO_4$. The mass of copper deposited at the cathode is: (at. mass of Cu = 63.5 amu)

(A) 63.5 g
(B) 2 g
(C) 127 g
(D) 0 g

Open solution

Q97. JEE Main 2014 (Online) 12th April Morning Slot

How many electrons would be required to deposit 6.35 g of copper at the cathode during the electrolysis of an aqueous solution of copper sulphate ? (Atomic mass of copper = 63.5 u, N_A = Avogadro's constant) :

(A) $\frac{N_A}{20}$
(B) $\frac{N_A}{10}$
(C) $\frac{N_A}{5}$
(D) $\frac{N_A}{2}$

Open solution

Q98. JEE Main 2014 (Online) 12th April Morning Slot

Copper becomes green when exposed to moist air for a long period. This is due to :

(A) the formation of a layer of cupric oxide on the surface of copper.
(B) the formation of a layer of basic carbonate of copper on the surface of copper.
(C) the formation of a layer of cupric hydroxide on the surface of copper.
(D) the formation of basic copper sulphate layer on the surface of the metal.

Open solution

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Q99. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

Given

$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$; $E^o = +0.77$ V

$Al^{3+}(aq) + 3e^- \rightarrow Al(s)$; $E^o = -1.66$ V

$Br_2(aq) + 2e^- \rightarrow 2Br^-$; $E^o = +1.09$ V

Considering the electrode potentials, which of the following represents the correct order of reducing power ?

(A) $Fe^{2+} < Al < Br^-$

(B) $Br^- < Fe^{2+} < Al$

(C) $Al < Br^- < Fe^{2+}$

(D) $Al < Fe^{2+} < Br^-$

Q100. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

The standard electrode potentials $\left(E^o_{M^+/M}\right)$ of four metals A, B, C and D are -1.2 V, 0.6 V, 0.85 V and -0.76 V, respectively. The sequence of deposition of metals on applying potential is :

(A) A, C, B, D

(B) B, D, C, A

(C) C, B, D, A

(D) D, A, B, C

Q101. JEE Main 2014 (Online) 9th April Morning Slot

Open solution

A current of 10.0 A flows for 2.00 h through an electrolytic cell containing a molten salt of metal X. This results in the deposition of 0.250 mol of metal X at the cathode. The oxidation state of X in the molten salt is : (F = 96,500 C)

(A) 1 +

(B) 2 +

(C) 3 +

(D) 4 +

Q102. JEE Main 2014 (Offline)

Open solution

Resistance of 0.2 M solution of an electrolyte is 50 Ω. The specific conductance of the solution is 1.4 S m^{-1}. The resistance of 0.5 M solution of the same electrolyte is 280 Ω. The molar conductivity of 0.5 M solution of the electrolyte in S m^2 mol^{-1} is :

(A) 5×10^3

(B) 5×10^2

(C) 5×10^{-4}

(D) 5×10^{-3}

Q103. JEE Main 2014 (Offline)

Open solution

Given below are the half-cell reactions:

$Mn^{2+} + 2e^- \rightarrow Mn$; $E^o = -1.18$ V

$2(Mn^{3+} + e^- \rightarrow Mn^{2+})$; $E^o = +1.51$ V

The E^o for $3Mn^{2+} \rightarrow Mn + 2Mn^{3+}$ will be :

(A) -0.33 V; the reaction will not occur

(B) -0.33 V; the reaction will occur

(C) -2.69 V; the reaction will not occur

(D) -2.69 V; the reaction will occur

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Q104. JEE Main 2014 (Offline)

The equivalent conductance of NaCl at concentration C and at infinite dilution are λ_C and λ_∞, respectively. The correct relationship between λ_C and λ_∞ is given as: (where the constant B is positive)

Open solution

(A) $\lambda_C = \lambda_\infty + (B)C$

(B) $\lambda_C = \lambda_\infty - (B)C$

(C) $\lambda_C = \lambda_\infty - (B)\sqrt{C}$

(D) $\lambda_C = \lambda_\infty + (B)\sqrt{C}$

Q105. JEE Main 2013 (Online) 25th April Morning Slot

The Gibbs energy for the decomposition of Al_2O_3 at 500°C is as follows :

$\frac{2}{3}Al_2O_3 \rightarrow \frac{4}{3}Al + O_2$, $\Delta_r G = +940$ kJ mol^{-1}

The potential difference needed for the electrolytic reduction of aluminium oxide at 500°C should be at least :

Open solution

(A) 4.5 V

(B) 3.0 V

(C) 5.0 V

(D) 2.5 V

Q106. JEE Main 2013 (Online) 25th April Morning Slot

A solution of copper sulphate ($CuSO_4$) is electrolysed for 10 minutes with a current of 1.5 amperes. The mass of copper deposited at the cathode (at. mass of Cu = 63u) is :

Open solution

(A) 0.3892 g

(B) 0.2938 g

(C) 0.2398 g

(D) 0.3928 g

Q107. JEE Main 2013 (Online) 23th April Morning Slot

Given :

$E^\circ_{\frac{1}{2}Cl_2/Cl^-} = 1.36$ V, $E^\circ_{Cr^{3+}/Cr} = -0.74$ V

$E^\circ_{Cr_2O_7^{2-}/Cr^{3+}} = 1.33$ V, $E^\circ_{MnO_4^-/Mn^{2+}} = 1.51$ V

The correct order of reducing power of the species (Cr, Cr^{3+}, Mn^{2+} and Cl^-) will be:

Open solution

(A) $Mn^{2+} < Cl^- < Cr^{3+} < Cr$

(B) $Mn^{2+} < Cr^{3+} < Cl^- < Cr$

(C) $Cr^{3+} < Cl^- < Mn^{2+} < Cr$

(D) $Cr^{3+} < Cl^- < Cr < Mn^{2+}$

Q108. JEE Main 2013 (Online) 9th April Morning Slot

Electrode potentials (E°) are given below :

$Cu^+/Cu = +0.52$ V,

$Fe^{3+}/Fe^{2+} = +0.77$ V,

$\frac{1}{2}I_2(s)/I^- = +0.54$ V,

$Ag^+/Ag = +0.88$ V.

Based on the above potentials, strongest oxidizing agent will be :

Open solution

(A) Cu^+

(B) Fe^{3+}

(C) Ag^+

(D) I_2

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Q109. JEE Main 2013 (Offline)

Open solution

Given

$E^o_{Cr^{2+}/Cr}$ = -0.74 V; $E^o_{MnO_4^-/Mn^{2+}}$ = 1.51 V

$E^o_{Cr_2O_7^{2-}/Cr^{3+}}$ = 1.33 V; E^o_{Cl/Cl^-} = 1.36 V

Based on the data given above, strongest oxidising agent will be :

(A) Cr^{3+} (B) Mn^{2+} (C) MnO_4^- (D) Cl^-

Q110. AIEEE 2012 (Online) 26th May

Open solution

Given

$E^\circ_{Cu^{2+}/Cu} = 0.34\ V,\ E^\circ_{Cu^{2+}/Cu^+} = 0.15\ V$

Standard electrode potential for the half cell Cu^+/Cu is

(A) $0.38\ V$ (B) $0.53\ V$ (C) $0.19\ V$ (D) $0.49\ V$

Q111. AIEEE 2012 (Online) 19th May

Open solution

The standard potentials of Ag^+/Ag, $Hg_2^{2+}/2Hg$, Cu^{2+}/Cu and Mg^{2+}/Mg electrodes are 0.80, 0.79, 0.34 and $-2.37\ V$, respectively. An aqueous solution which contains one mole per litre of the salts of each of the four metals is electrolyzed. With increasing voltage, the correct sequence of deposition of the metals at the cathode is

(A) Ag, Hg, Cu, Mg (B) Cu, Hg, Ag only (C) Ag, Hg, Cu only (D) Mg, Cu, Hg, Ag

Q112. AIEEE 2012 (Online) 7th May

Open solution

A battery is constructed of Cr and $Na_2Cr_2O_7$. The unbalanced chemical equation when such a battery discharges is following:

$Na_2Cr_2O_7 + Cr + H^+ \rightarrow Cr^{3+} + H_2O + Na^+$

If one Faraday of electricity is passed through the battery during the charging, the number of moles of Cr^{3+} removed from the solution is

(A) $\frac{4}{3}$ (B) $\frac{1}{3}$ (C) $\frac{3}{3}$ (D) $\frac{2}{3}$

Q113. AIEEE 2012

Open solution

The standard reduction potentials for Zn^{2+}/ Zn, Ni^{2+}/ Ni, and Fe^{2+}/ Fe are −0.76, −0.23 and −0.44 V respectively. The reaction

$X + Y^{2+} \rightarrow X^{2+} + Y$ will be spontaneous when :

(A) X = Ni, Y = Fe (B) X = Ni, Y = Zn (C) X = Fe, Y = Zn (D) X = Zn, Y = Ni

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Q114. AIEEE 2011

The reduction potential of hydrogen half cell will be negative if :

(A) $p(H_2)$ = 1 atm and $[H^+]$ = 1.0 M

(B) $p(H_2)$ = 1 atm and $[H^+]$ = 2.0 M

(C) $p(H_2)$ = 2 atm and $[H^+]$ =1.0 M

(D) $p(H_2)$ = 2 atm and $[H^+]$ =2.0 M

Open solution

Q115. AIEEE 2010

The Gibbs energy for the decomposition of Al_2O_3 at 500°C is as follows :

$\frac{2}{3}Al_2O_3 \rightarrow \frac{4}{3}Al + O_2$, $\Delta_r G$ = + 966 kJ mol^{-1}

The potential difference needed for electrolytic reduction of Al_2O_3 at 500°C is at least :

(A) 4.5 V (B) 3.0 V (C) 2.5 V (D) 5.0 V

Open solution

Q116. AIEEE 2009

In a fuel cell methanol is used as fuel and oxygen gas is used as an oxidizer. The reaction is $CH_3OH(l) + 3/2O_2 \rightarrow CO_2(g) + 2H_2O(l)$

At 298K standard Gibb's energies of formation for $CH_3OH(l)$, $H_2O(l)$ and $CO_2(g)$ are -166.2, -237.2 and -394.4 kJ mol^{-1} respectively. If standard enthalpy of combustion of methanol is -726 kJ mol^{-1}, efficiency of the fuel cell will be

(A) 87% (B) 90% (C) 97% (D) 80%

Open solution

Q117. AIEEE 2009

Given : $E^o_{Fe^{3+}/Fe}$ = -0.036V; $E^o_{Fe^{2+}/Fe}$ = -0.439 V

The value of standard electrode potential for the change,

$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$ will be

(A) -0.072 V (B) 0.385 V (C) 0.770 V (D) -0.270 V

Open solution

Q118. AIEEE 2008

Given $E^o_{Cr^{3+}/Cr}$ = -0.72 V; $E^o_{Fe^{2+}/Fe}$ = -0.42 V. The potential for the cell Cr | Cr^{3+} (0.1M) || Fe^{2+} (0.01 M) | Fe is

(A) 0.26 V (B) 0.399 V (C) −0.339 V (D) −0.26 V

Open solution

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Q119. AIEEE 2007

Open solution

The equivalent conductances of two strong electrolytes at infinite dilution in H_2O (where ions move freely through a solution) at 25°C are given below:

$\wedge^o_{CH_3COONa}$ = 91.0 S cm^2/equiv

$\wedge^o_{HCl}$ = 426.2 S cm^2/equiv

What additional information/quantity one needs to calculate $\wedge^o$ of an aqueous solution of acetic acid?

(A) $\wedge^o$ of chloroacetic acid (C/CH_2COOH)

(B) $\wedge^o$ of NaCl

(C) $\wedge^o$ of CH_3COOK

(D) The limiting equivalent conductance of $H^+(\wedge^o_{H^+})$

Q120. AIEEE 2007

Open solution

The cell, Zn | Zn^{2+} (1M) || Cu^{2+} (1M) | Cu(E^o_{cell} = 1.10V) was allowed to be completely discharged at 298 K. The relative concentration of Zn^{2+} to Cu^{2+} $\left[\frac{[Zn^{2+}]}{[Cu^{2+}]}\right]$ is

(A) antilog (24.08) (B) 37.3 (C) $10^{37.3}$ (D) 9.65×10^4

Q121. AIEEE 2006

Open solution

Given the data at 25°C,

$Ag + I^- \rightarrow AgI + e^-$, E^o = 0.152 V

$Ag \rightarrow Ag^+ + e^-$, E^o = -0.800 V

What is the value of log K_{sp} for AgI? (2.303 RT/F = 0.059 V)

(A) −8.12 (B) +8.612 (C) −37.83 (D) −16.13

Q122. AIEEE 2006

Open solution

The molar conductivities $\wedge^o_{NaOAc}$ and $\wedge^o_{HCl}$ and at infinite dilution in water at 25°C are 91.0 and 426.2 Scm^2/mol respectively. To calculate $\wedge^o_{HOAc}$, the additional value required is

(A) $\wedge^o_{H_2O}$ (B) $\wedge^o_{KCl}$ (C) $\wedge^o_{NaOH}$ (D) $\wedge^o_{NaCl}$

Q123. AIEEE 2006

Open solution

Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1 M is 100Ω. The conductivity of this solution is 1.29 S m^{-1}. Resistance of the same cell when filled with 0.2 M of the same solution is 520 Ω. The molar conductivity of 0.2 M solution of the electrolyte will be

(A) 124×10^{-4} S m^2 mol^{-1} (B) 1240×10^{-4} S m^2 mol^{-1}

(C) 1.24×10^{-4} S m^2 mol^{-1} (D) 12.4×10^{-4} S m^2 mol^{-1}

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Q124. AIEEE 2005

For a spontaneous reaction the ΔG , equilibrium constant (K) and E^o_{cell} will be respectively

(A) -ve, >1, +ve (B) +ve, >1, -ve (C) -ve, <1, -ve (D) -ve, >1, -ve

Open solution

Q125. AIEEE 2005

The highest electrical conductivity of the following aqueous solutions is of :

(A) 0.1 M acetic acid (B) 0.1 M chloroacetic acid

(C) 0.1 M fluoroacetic acid (D) 0.1 M difluoroacetic acid

Open solution

Q126. AIEEE 2005

Aluminium oxide may be electrolysed at 1000°C to furnish aluminium metal (Atomic mass = 27 amu; 1 Faraday = 96,500 Coulombs). The cathode reaction is $Al^{3+} + 3e^- \rightarrow Al^o$ To prepare 5.12 kg of aluminium metal by this method would require

(A) 5.49×10^7 C of electricity (B) 1.83×10^7 C of electricity

(C) 5.49×10^4 C of electricity (D) 5.49×10^1 C of electricity

Open solution

Q127. AIEEE 2005

Electrolyte:	KCl	KNO_3	HCl	NaOAc	NaCl
$\wedge^\infty (Scm^2mol^{-1})$:	149.9	145	426.2	91	126.5

Calculate $\wedge^\infty_{HOAc}$ using appropriate molar conductances of the electrolytes listed above at infinite dilution in H_2O at 25°C

(A) 517.2 (B) 552.7 (C) 390.7 (D) 217.5

Open solution

Q128. AIEEE 2004

In a hydrogen – oxygen fuel cell, combustion of hydrogen occurs to :

(A) generate heat

(B) remove adsorbed oxygen from electrode surfaces

(C) produce high purity water

(D) create potential difference between the two electrodes

Open solution

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Q129. AIEEE 2004

Open solution

Consider the following E^o values

$E^o_{Fe^{3+}/Fe^{2+}}$ = 0.77 V;

$E^o_{Sn^{2+}/Sn}$ = -0.14 V

Under standard conditions the potential for the reaction

$Sn(s) + 2Fe^{3+}(aq) \rightarrow 2Fe^{2+}(aq) + Sn^{2+}(aq)$ is :

(A) 1.68 V (B) 0.63 V (C) 0.91 V (D) 1.40 V

Q130. AIEEE 2004

Open solution

The standard e.m.f of a cell, involving one electron change is found to be 0.591 V at 25°C. The equilibrium constant of the reaction is (F = 96,500 C mol^{-1}: R = 8.314 JK^{-1} mol^{-1})

(A) 1.0×10^1 (B) 1.0×10^{30} (C) 1.0×10^{10} (D) 1.0×10^5

Q131. AIEEE 2004

Open solution

The limiting molar conductivities Λ° for NaCl, KBr and KCl are 126, 152 and 150 S cm^2 mol^{-1} respectively. The Λ° for NaBr is

(A) 128 S cm^2 mol^{-1} (B) 278 S cm^2 mol^{-1} (C) 176 S cm^2 mol^{-1} (D) 302 S cm^2 mol^{-1}

Q132. AIEEE 2004

Open solution

In a cell that utilises the reaction Zn(s) + $2H^+$ (aq) $\rightarrow$ Zn2+(aq) + H2(g) addition of H_2SO_4 to cathode compartment, will

(A) lower the E and shift equilibrium to the left

(B) increases the E and shift equilibrium to the left

(C) increase the E and shift equilibrium to the right

(D) Lower the E and shift equilibrium to the right

Q133. AIEEE 2004

Open solution

The $E^o_{M^{3+}/M^{2+}}$ values for Cr, Mn, Fe and Co are – 0.41, +1.57, + 0.77 and +1.97 V respectively. For which one of these metals the change in oxidation state from +2 to +3 is easiest?

(A) Fe (B) Mn (C) Cr (D) Co

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Q134. AIEEE 2003

For a cell reaction involving a two-electron change, the standard e.m.f. of the cell is found to be 0.295 V at 25°C. The equilibrium constant of the reaction at 25°C will be

(A) 29.5×10^{-2} (B) 10 (C) 1×10^{10} (D) 1×10^{-10}

Open solution

Q135. AIEEE 2003

Standard reduction electrode potentials of three metals A,B&C are respectively +0.5 V, -3.0 V & -1.2 V. The reducing, powers of these metals are

(A) A > B > C (B) C > B > A (C) A > C > B (D) B > C > A

Open solution

Q136. AIEEE 2003

When during electrolysis of a solution of $AgNO_3$, 9650 coulombs of charge pass through the electroplating bath, the mass of silver deposited on the cathode will be :

(A) 10.8 g (B) 21.6 g (C) 108 g (D) 1.08 g

Open solution

Q137. AIEEE 2003

For the redox reaction Zn(s) + Cu^{2+}(0.1 M) $\rightarrow$ Zn^{2+}(1M) + Cu(s) taking place in a cell, E^o_{cell} is 1.10 volt. E_{cell} for the cell will be ($2.303\frac{RT}{F}$ = 0.0591)

(A) 1.80 volt (B) 1.07 volt (C) 0.82 volt (D) 2.14 volt

Open solution

Q138. AIEEE 2003

Several blocks of magnesium are fixed to the bottom of a ship to :

(A) make the ship lighter (B) prevent action of water and salt

(C) prevent puncturing by under-sea rocks (D) keep away the sharks

Open solution

Q139. AIEEE 2002

Conductivity (Siemens S) is directly proportional to area of the vessel and the concentration of the solution in it and is inversely proportional to the length of the vessel, then the constant of proportionality is expressed in :

(A) Sm mol^{-1} (B) Sm^2 mol^{-1} (C) $S^{-2}m^2$ mol (D) S^2m^2 mol^{-2}

Open solution

Q140. AIEEE 2002

EMF of a cell in terms of reduction potential of its left and right electrodes is :

(A) $E = E_{left} - E_{right}$ (B) $E = E_{left} + E_{right}$

(C) $E = E_{right} - E_{left}$ (D) $E = -(E_{right} + E_{left})$

Open solution

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Q141. AIEEE 2002

For the following cell with hydrogen electrodes at two different pressures p_1 and p_2. What will be the emf for the given cell :

$$\underset{p_1}{Pt(H_2)}|\underset{1M}{H^+(aq)}|\underset{p_2}{Pt(H_2)}$$

(A) $\frac{RT}{F}\log_e\frac{P_1}{P_2}$

(B) $\frac{RT}{2F}\log_e\frac{P_1}{P_2}$

(C) $\frac{RT}{F}\log_e\frac{P_2}{P_1}$

(D) none of these

Open solution

Q142. AIEEE 2002

Which of the following reaction is possible at anode?

(A) $2Cr^{3+} + 7H_2O \rightarrow Cr_2O_7^{2-} + 14H^+$

(B) $F_2 \rightarrow 2F^-$

(C) $(1/2)\ O_2 + 2H^+ \rightarrow H_2O$

(D) none of these

Open solution

Q143. AIEEE 2002

When the sample of copper with zinc impurity is to be purified by electrolysis, the appropriate electrodes are :

(A) cathode = pure zinc, anode = pure copper

(B) cathode = impure sample, anode = pure copper

(C) cathode = impure zinc, anode = impure sample

(D) cathode = pure copper, anode = impure sample

Open solution

Q144. AIEEE 2002

For a cell given below

$$\underset{-}{Ag|Ag^+}||\underset{+}{Cu^{2+}|Cu}$$

$Ag^+ + e^- \longrightarrow Ag; E^\circ = x$

$Cu^{2+} + 2e^- \longrightarrow Cu; E^\circ = y$

E° cell is :

(A) $x + 2y$

(B) $2x + y$

(C) $y - x$

(D) $y - 2x$

Open solution

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Numerical 92 questions

Q145. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

At 298 K , the molar conductivity of $x\%(\mathrm{w/w})\mathrm{MX}$ solution (aqueous) is $123.5\ \mathrm{S\ cm^2\ mol^{-1}}$. The conductance of same solution is $1.9 \times 10^{-3}\ \mathrm{S}$. The value of x is ____ $\times 10^{-2}$.

(Given: cell constant $= 1.3\ \mathrm{cm^{-1}}$; molar mass of MX is $75\ \mathrm{g\ mol^{-1}}$, density of aqueous solution of MX at 298 K is $1.0\ \mathrm{g\ mL^{-1}}$)

Q146. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

An electrochemical cell consists of the following two redox couples,
$\mathrm{M}^{x+}(\mathrm{aq})/\mathrm{M}(\mathrm{s})\ \left[\mathrm{E}^{\Theta}_{\mathrm{red}} = +0.15\ \mathrm{V}\right]$ and $\mathrm{Fe}^{3+}(\mathrm{aq})/\mathrm{Fe}(\mathrm{s})\ \left[\mathrm{E}^{\Theta}_{\mathrm{red}} = -0.036\ \mathrm{V}\right]$. The cell EMF $\left(\mathrm{E}_{\mathrm{cell}}\right)$ is recorded to be 0.2057 V . If the reaction quotient of the electrochemical reaction is found to be 10^{-2}, then the value of x is

____ .(Nearest integer)

[Given : M is a p-block metal and $\frac{2.303RT}{F} = 0.059\ \mathrm{V}$]

Q147. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

Consider the following two half-cell reactions along with the standard reduction potential given :

$\mathrm{CO_2 + 6H^+ + 6}e^- \longrightarrow \mathrm{CH_3OH + H_2O} \qquad E^{\circ}_{\mathrm{red}} = 0.02\ \mathrm{V}$

$\frac{1}{2}\mathrm{O_2 + 2H^+ + 2}e^- \longrightarrow \mathrm{H_2O} \qquad E^{\circ}_{\mathrm{red}} = 1.23\ \mathrm{V}$

A fuel cell was set up using the above two reactions such that the cell operates under the standard condition of 1 bar pressure and 298 K temperature. The fuel cell works with 80% efficiency. If the work derived from the cell using 1 mol of $\mathrm{CH_3OH}$ is used to compress an ideal gas isothermally against a constant pressure of 1 kPa, then the change in the volume of the gas,

$\Delta V =$ _____ $\mathrm{m^3}$. (nearest integer)

Given : $\mathrm{F} = 96500\ \mathrm{C\ mol^{-1}}$

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Q148. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

A volume of x mL of 5 M $NaHCO_3$ solution was mixed with 10 mL of 2 M H_2CO_3 solution to make an electrolytic buffer. If the same buffer was used in the following electrochemical cell to record a cell potential of 235.3 mV, then the value of **x** = _____ mL (nearest integer).

$Sn(s) \mid Sn(OH)_6^{2-}$ (0.5 M) $\mid HSnO_2^-$ (0.05 M) $\mid OH^- \mid Bi_2O_3(s) \mid Bi(s)$

Consider up to one place of decimal for intermediate calculations

Given: $E^o_{Sn(OH)_6^{2-}|HSnO_2^-} = -0.9V$

$E^o_{Bi_2O_3|Bi} = -0.44\ V$

$pKa_{(H_2CO_3)} = 6.11$

$\frac{2.303RT}{F} = 0.059\ V$

Antilog $(1.29) = 19.5$

Q149. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

For strong electrolyte Λ_m increases slowly with dilution and can be represented by the equation

$\Lambda_m = \Lambda_m^\circ - Ac^{1/2}$

Molar conductivity values of the solutions of strong electrolyte AB at 18°C are given below :

c [mol L^{-1}]	0.04	0.09	0.16	0.25
Λ_m [S cm^2 mol^{-1}]	96.1	95.7	95.3	94.9

The value of constant A based on the above data [in S cm^2 mol^{-1}/(mol/L)$^{1/2}$] unit is _____.

Q150. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

Consider the following redox reaction taking place in acidic medium

$BH_4^-(aq) + ClO_3^-(aq) \longrightarrow H_2BO_3^-(aq) + Cl^-(aq)$

If the Nernst equation for the above balanced reaction is

$E_{cell} = E^\circ_{cell} - \frac{RT}{nF}\ln Q$,

then the value of n is ____ .(Nearest integer)

Q151. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

Molar conductivity of a weak acid HQ of concentration 0.18 M was found to be $1/30$ of the molar conductivity of another weak acid HZ with concentration of 0.02 M . If $\lambda^\circ Q^-$ happened to be equal with $\lambda^\circ Z^-$, then the difference of the pK_a values of the two weak acids $(pK_a(HQ) - pK_a(HZ))$ is ____ (Nearest integer).

[Given: degree of dissociation $(\alpha) \ll 1$ for both weak acids, λ° : limiting molar conductivity of ions]

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Q152. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Electricity is passed through an acidic solution of Cu^{2+} till all the Cu^{2+} was exhausted, leading to the deposition of 300 mg of Cu metal. However, a current of 600 mA was continued to pass through the same solution for another 28 minutes by keeping the total volume of the solution fixed at 200 mL . The total volume of oxygen evolved at STP during the entire process is ____ mL . (Nearest integer)

[Given:

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s) E^{\circ}_{red} = +0.34\ V$

$O_2(g) + 4H^+ + 4e^- \rightarrow 2H_2O E^{\circ}_{red} = +1.23\ V$

Molar mass of $Cu = 63.54\ g\ mol^{-1}$

Molar mass of $O_2 = 32\ g\ mol^{-1}$

Faraday Constant $= 96500 C mol^{-1}$

Molar volume at $STP = 22.4\ L$]

Q153. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Consider the following electrochemical cell :

$Pt\,|O_2(g)(1bar)|HCl(aq)||M^{2+}(aq, 1.0M)\,|\,M(s)$

The pH above which, oxygen gas would start to evolve at anode is ____ (nearest integer).

$$\left[\text{Given : } \left.\begin{matrix} E^{\circ}_{M^{2+}/M} = 0.994\ V \\ E^{\circ}_{O_2/H_2O} = 1.23\ V \end{matrix}\right\} \text{ standard reduction potential and } \frac{RT}{F}(2.303) = 0.059\ V \text{ at the given condition}\right]$$

Q154. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Consider the following electrochemical cell at 298 K

$Pt\,|HSnO_2^-(aq)|Sn(OH)_6{}^{2-}(aq)\,|OH^-(aq)|Bi_2O_3(s)\,|\,Bi(s).$

If the reaction quotient at a given time is 10^6, then the cell EMF (E_{cell}) is

____ $\times 10^{-1}$ V (Nearest integer).

Given the standard half-cell reduction potential as

$E^{\circ}_{Bi_2O_3/Bi,OH^-} = -0.44\ V$ and $E^{\circ}_{Sn(OH)_6^{2-}/HSnO_2^-,OH^-} = -0.90\ V$

Q155. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

MX is a sparingly soluble salt that follows the given solubility equilibrium at 298 K.

$MX(s) \rightleftharpoons M^+(aq) + X^-(aq); \quad K_{sp} = 10^{-10}$

If the standard reduction potential for M^+ (aq) + e^- → M(s) is $\left(E^{\ominus}_{M^+/M}\right) = 0.79$ V, then the value of the standard reduction potential for the metal/metal insoluble salt electrode $E^{\ominus}_{X^-/MX(s)/M}$ is _____ mV. (nearest integer)

[Given: $\frac{2.303RT}{F} = 0.059$ V]

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Q156. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

The pH and conductance of a weak acid (HX) was found to be 5 and 4×10^{-5} S, respectively. The conductance was measured under standard condition using a cell where the electrode plates having a surface area of $1\ cm^2$ were at a distance of 15 cm apart. The value of the limiting molar conductivity is ____ $Sm^2\ mol^{-1}$. (nearest integer)

(Given : degree of dissociation of the weak acid $(\alpha) \ll 1$)

Q157. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

Consider the following half cell reaction

$Cr_2O_7^{2-}(aq) + 6e^- + 14H^+(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(\ell)$

The reaction was conducted with the ratio of $\frac{[Cr^{3+}]^2}{[Cr_2O_7^{2-}]} = 10^{-6}$. The pH value at which the EMF of the half cell will become zero is _____.

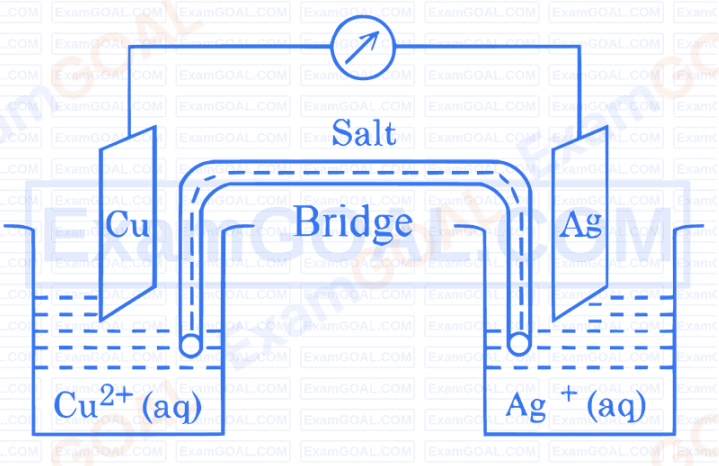
(nearest integer value)

[Given: standard half cell reduction potential $E^\circ_{Cr_2O_7^{2-},H^+/Cr^{3+}} = 1.33$ V, $\frac{2.303RT}{F} = 0.059$ V.]

Q158. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

1 Faraday electricity was passed through $Cu^{2+}(1.5M, 1\ L)/Cu$ and 0.1 Faraday was passed through $Ag^+(0.2M, 1\ L)/Ag$ electrolytic cells. After this the two cells were connected as shown below to make an electrochemical cell. The emf of the cell thus formed at 298 K is _____ mV (nearest integer)



Given : $E^\circ Cu^{2+}/Cu = 0.34\ V$

$E^\circ Ag^+/Ag = 0.8\ V$

$\frac{2.303RT}{F} = 0.06\ V$

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Q159. JEE Main 2025 (Online) 4th April Evening Shift

The molar conductance of an infinitely dilute solution of ammonium chloride was found to be $185\ \mathrm{S\ cm^2\ mol^{-1}}$ and the ionic conductance of hydroxyl and chloride ions are 170 and $70\ \mathrm{S\ cm^2\ mol^{-1}}$, respectively. If molar conductance of 0.02 M solution of ammonium hydroxide is $85.5\ \mathrm{S\ cm^2\ mol^{-1}}$, its degree of dissociation is given by $x \times 10^{-1}$. The value of x is _____ . (Nearest integer)

Open solution

Q160. JEE Main 2025 (Online) 2nd April Evening Shift

$0.2\%(\mathrm{w/v})$ solution of NaOH is measured to have resistivity $870.0\ \mathrm{m\Omega\ m}$. The molar conductivity of the solution will be_____$\times 10^2 \mathrm{mSdm^2\ mol^{-1}}$. (Nearest integer)

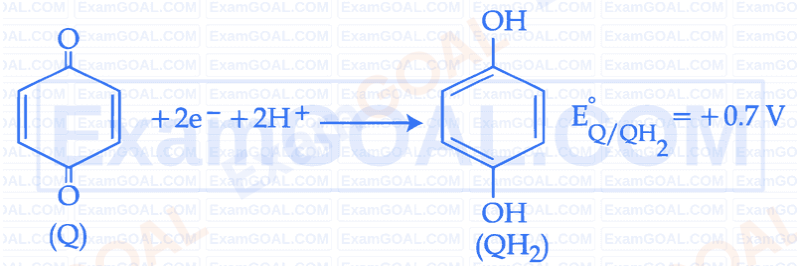
Open solution

Q161. JEE Main 2025 (Online) 2nd April Morning Shift

Consider the following electrochemical cell at standard condition.

$\mathrm{Au(s)\,|QH_2, Q|NH_4X(0.01M)||Ag^+(1M)\,|\,Ag(s)}\quad \mathrm{E_{cell}} = +0.4\ \mathrm{V}$

The couple $\mathrm{QH_2/Q}$ represents quinhydrone electrode, the half cell reaction is given below:



$\left[\text{Given : } \mathrm{E^0_{Ag^+/Ag}} = +0.8\ \mathrm{V} \text{ and } \frac{2.303\mathrm{RT}}{\mathrm{F}} = 0.06\ \mathrm{V}\right]$

The $\mathrm{pK_b}$ value of the ammonium halide salt $(\mathrm{NH_4X})$ used here is _____ . (nearest integer)

Open solution

Q162. JEE Main 2025 (Online) 28th January Evening Shift

Electrolysis of 600 mL aqueous solution of NaCl for 5 min changes the pH of the solution to 12 .

The current in Amperes used for the given electrolysis is _____ . (Nearest integer).

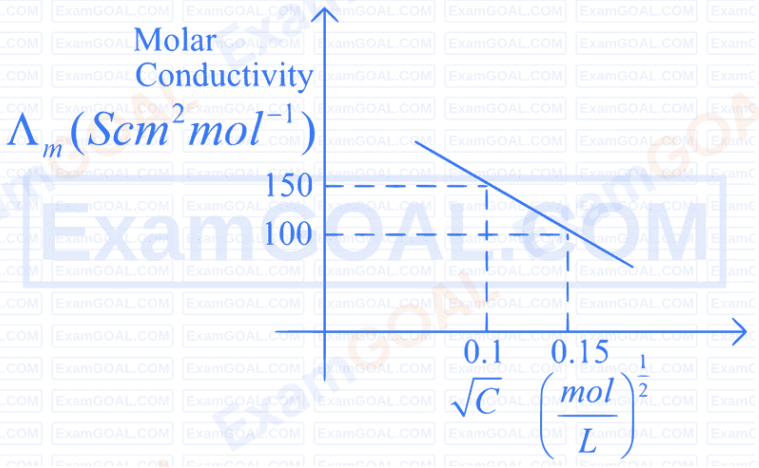
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Q163. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Given below is the plot of the molar conductivity vs $\sqrt{\text{concentration}}$ for KCl in aqueous solution.



If, for the higher concentration of KCl solution, the resistance of the conductivity cell is 100Ω, then the resistance of the same cell with the dilute solution is ' x ' Ω

The value of x is _____ (Nearest integer)

Q164. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

The standard reduction potentials at $298\ \text{K}$ for the following half cells are given below :

$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}, \quad \text{E}^\circ = 1.33\ \text{V}$

$\text{Fe}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Fe} \quad \text{E}^\circ = -0.04\ \text{V}$

$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ni} \quad \text{E}^\circ = -0.25\ \text{V}$

$\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag} \quad \text{E}^\circ = 0.80\ \text{V}$

$\text{Au}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Au} \quad \text{E}^\circ = 1.40\ \text{V}$

Consider the given electrochemical reactions,

The number of metal(s) which will be oxidized by $\text{Cr}_2\text{O}_7^{2-}$, in aqueous solution is _____.

Q165. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

The amount of electricity in Coulomb required for the oxidation of $1\ \text{mol}$ of H_2O to O_2 is _____ $\times 10^5\text{C}$.

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Q166. JEE Main 2024 (Online) 1st February Evening Shift

Consider the following redox reaction :

$$MnO_4^- + H^+ + H_2C_2O_4 \rightleftharpoons Mn^{2+} + H_2O + CO_2$$

The standard reduction potentials are given as below $\left(E^0_{red}\right)$:

$E^\circ_{MnO_4^-/Mn^{2+}} = +1.51\ V$

$E^\circ_{CO_2/H_2C_2O_4} = -0.49\ V$

If the equilibrium constant of the above reaction is given as $K_{eq} = 10^x$, then the value of $x =$ _____ (nearest integer)

Open solution

Q167. JEE Main 2024 (Online) 1st February Morning Shift

The potential for the given half cell at $298\ K$ is (-) _____ $\times 10^{-2}\ V$

$$2H^+_{(aq)} + 2e^- \longrightarrow H_2(g)$$

$[H^+] = 1M, P_{H_2} = 2\ atm$

(Given : $2.303RT/F = 0.06\ V, \log 2 = 0.3$)

Open solution

Q168. JEE Main 2024 (Online) 31st January Evening Shift

The values of conductivity of some materials at $298.15\ K$ in Sm^{-1} are 2.1×10^3,

$1.0 \times 10^{-16}, 1.2 \times 10, 3.91, 1.5 \times 10^{-2}, 1 \times 10^{-7}, 1.0 \times 10^3$.

The number of conductors among the materials is _____.

Open solution

Q169. JEE Main 2024 (Online) 31st January Morning Shift

One Faraday of electricity liberates $x \times 10^{-1}$ gram atom of copper from copper sulphate. x is _____.

Open solution

Q170. JEE Main 2024 (Online) 29th January Evening Shift

A constant current was passed through a solution of $AuCl_4^-$ ion between gold electrodes. After a period of 10.0 minutes, the increase in mass of cathode was $1.314\ g$. The total charge passed through the solution is _____ $\times 10^{-2}\ F$.

(Given atomic mass of $Au = 197$)

Open solution

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Q171. JEE Main 2024 (Online) 29th January Morning Shift

The mass of zinc produced by the electrolysis of zinc sulphate solution with a steady current of $0.015\ \mathrm{A}$ for 15 minutes is _____ $\times 10^{-4}$ g.

(Atomic mass of zinc $= 65.4\ \mathrm{amu}$)

Open solution

Q172. JEE Main 2024 (Online) 27th January Evening Shift

The hydrogen electrode is dipped in a solution of $\mathrm{pH} = 3$ at $25^{\circ}\mathrm{C}$. The potential of the electrode will be _____ $\times 10^{-2}$ V.

$\left(\frac{2.303\mathrm{RT}}{\mathrm{F}} = 0.059\ \mathrm{V}\right)$

Open solution

Q173. JEE Main 2024 (Online) 27th January Morning Shift

The mass of silver (Molar mass of $\mathrm{Ag}: 108\ \mathrm{gmol}^{-1}$) displaced by a quantity of electricity which displaces $5600\ \mathrm{mL}$ of O_2 at S.T.P. will be _____ g.

Open solution

Q174. JEE Main 2023 (Online) 15th April Morning Shift

The number of correct statements from the following is _____.

(A) Conductivity always decreases with decrease in concentration for both strong and weak electrolytes.

(B) The number of ions per unit volume that carry current in a solution increases on dilution.

(C) Molar conductivity increases with decrease in concentration

(D) The variation in molar conductivity is different for strong and weak electrolytes

(E) For weak electrolytes, the change in molar conductivity with dilution is due to decrease in degree of dissociation.

Open solution

Q175. JEE Main 2023 (Online) 13th April Evening Shift

At $298\ \mathrm{K}$, the standard reduction potential for $\mathrm{Cu}^{2+}/\mathrm{Cu}$ electrode is $0.34\ \mathrm{V}$.

Given : $\mathrm{K}_{\mathrm{sp}}\mathrm{Cu(OH)}_2 = 1 \times 10^{-20}$

Take $\frac{2.303\mathrm{RT}}{\mathrm{F}} = 0.059\ \mathrm{V}$

The reduction potential at $\mathrm{pH} = 14$ for the above couple is $(-)x \times 10^{-2}$ V. The value of x is _____

Open solution

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Q176. JEE Main 2023 (Online) 13th April Morning Shift

A metal surface of $100\ \mathrm{cm}^2$ area has to be coated with nickel layer of thickness $0.001\ \mathrm{mm}$. A current of $2\ \mathrm{A}$ was passed through a solution of $\mathrm{Ni(NO_3)_2}$ for 'x' seconds to coat the desired layer. The value of x is _____. (Nearest integer) (ρ_{Ni} (density of Nickel) is $10\ \mathrm{g\ mL^{-1}}$, Molar mass of Nickel is $60\ \mathrm{g\ mol^{-1}}$ $\mathrm{F} = 96500\ \mathrm{C\ mol^{-1}}$)

Open solution

Q177. JEE Main 2023 (Online) 11th April Evening Shift

The number of correct statements from the following is _____

A. $\mathrm{E_{cell}}$ is an intensive parameter

B. A negative $\mathrm{E^{\ominus}}$ means that the redox couple is a stronger reducing agent than the $\mathrm{H^+/H_2}$ couple.

C. The amount of electricity required for oxidation or reduction depends on the stoichiometry of the electrode reaction.

D. The amount of chemical reaction which occurs at any electrode during electrolysis by a current is proportional to the quantity of electricity passed through the electrolyte.

Open solution

Q178. JEE Main 2023 (Online) 11th April Morning Shift

In an electrochemical reaction of lead, at standard temperature, if $\mathrm{E^0\left(Pb^{2+}/Pb\right)} = \mathrm{m}$ Volt and $\mathrm{E^0\left(Pb^{4+}/Pb\right)} = \mathrm{n}$ Volt, then the value of $\mathrm{E^0\left(Pb^{2+}/Pb^{4+}\right)}$ is given by $\mathrm{m - xn}$. The value of x is _____. (Nearest integer)

Open solution

Q179. JEE Main 2023 (Online) 10th April Evening Shift

The specific conductance of $0.0025\ \mathrm{M}$ acetic acid is $5 \times 10^{-5}\ \mathrm{S\ cm^{-1}}$ at a certain temperature. The dissociation constant of acetic acid is _____ $\times 10^{-7}$ (Nearest integer)

Consider limiting molar conductivity of $\mathrm{CH_3COOH}$ as $400\ \mathrm{S\ cm^2\ mol^{-1}}$

Open solution

Q180. JEE Main 2023 (Online) 10th April Morning Shift

$\mathrm{FeO_4^{2-} \xrightarrow{+2.2V} Fe^{3+} \xrightarrow{+0.70V} Fe^{2+} \xrightarrow{-0.45V} Fe^0}$

$E^{\theta}_{FeO_4^{2-}/Fe^{2+}}$ is $x \times 10^{-3}$ V. The value of x is _____

Open solution

Q181. JEE Main 2023 (Online) 8th April Evening Shift

The number of incorrect statements from the following is _____.

A. The electrical work that a reaction can perform at constant pressure and temperature is equal to the reaction Gibbs energy.

B. $\mathrm{E^{\circ}_{cell}}$ cell is dependent on the pressure.

C. $\frac{dE^{\theta}\ \mathrm{cell}}{\mathrm{dT}} = \frac{\Delta_r S^{\theta}}{\mathrm{nF}}$

D. A cell is operating reversibly if the cell potential is exactly balanced by an opposing source of potential difference.

Open solution

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Q182. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

The standard reduction potentials at $298\ \mathrm{K}$ for the following half cells are given below:

$\mathrm{NO_3^- + 4H^+ + 3e^- \rightarrow NO(g) + 2H_2O} \quad \mathrm{E^\theta = 0.97\ V}$

$\mathrm{V^{2+}(aq) + 2e^- \rightarrow V} \qquad \mathrm{E^\theta = -1.19\ V}$

$\mathrm{Fe^{3+}(aq) + 3e^- \rightarrow Fe} \qquad \mathrm{E^\theta = -0.04\ V}$

$\mathrm{Ag^+(aq) + e^- \rightarrow Ag(s)} \qquad \mathrm{E^\theta = 0.80\ V}$

$\mathrm{Au^{3+}(aq) + 3e^- \rightarrow Au(s)} \qquad \mathrm{E^\theta = 1.40\ V}$

The number of metal(s) which will be oxidized by $\mathrm{NO_3^-}$ in aqueous solution is _____.

Q183. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

$1 \times 10^{-5}\ \mathrm{M\ AgNO_3}$ is added to $1\ \mathrm{L}$ of saturated solution of AgBr. The conductivity of this solution at $298\ \mathrm{K}$ is _____ $\times 10^{-8}\ \mathrm{S\ m^{-1}}$. (Nearest integer)

[Given : $\mathrm{K_{SP}(AgBr)} = 4.9 \times 10^{-13}$ at $298\ \mathrm{K}$

$\lambda^0_{\mathrm{Ag^+}} = 6 \times 10^{-3}\ \mathrm{S\ m^2\ mol^{-1}}$

$\lambda^0_{\mathrm{Br^-}} = 8 \times 10^{-3}\ \mathrm{S\ m^2\ mol^{-1}}$

$\lambda^0_{\mathrm{NO_3^-}} = 7 \times 10^{-3}\ \mathrm{S\ m^2\ mol^{-1}}$]

Q184. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

At what pH, given half cell $\mathrm{MnO_4^-(0.1\ M)\ |\ Mn^{2+}(0.001\ M)}$ will have electrode potential of 1.282 V? _____ (Nearest Integer)

Given $\mathrm{E^\circ_{MnO_4^-|Mn^{2+}}} = 1.54\ \mathrm{V}$, $\frac{2.303\mathrm{RT}}{\mathrm{F}} = 0.059\mathrm{V}$

Q185. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

The resistivity of a $0.8\mathrm{M}$ solution of an electrolyte is $5 \times 10^{-3}\Omega\ \mathrm{cm}$.

Its molar conductivity is _____ $\times 10^4\ \Omega^{-1}\ \mathrm{cm^2\ mol^{-1}}$. (Nearest integer)

Q186. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

The logarithm of equilibrium constant for the reaction $\mathrm{Pd^{2+} + 4Cl^- \rightleftharpoons PdCl_4^{2-}}$ is _____ (Nearest integer)

Given : $\frac{2.303RT}{\mathrm{F}} = 0.06\ \mathrm{V}$

$\mathrm{Pd^{2+}_{(aq)} + 2e^- \rightleftharpoons Pd(s)} \quad \mathrm{E^\ominus = 0.83\ V}$

$\mathrm{PdCl_4^{2-}(aq) + 2e^- \rightleftharpoons Pd(s) + 4Cl^-(aq)} \mathrm{E^\ominus = 0.65\ V}$

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Q187. JEE Main 2023 (Online) 30th January Evening Shift

The electrode potential of the following half cell at $298\ \mathrm{K}$

$\mathrm{X}\left|\mathrm{X}^{2+}(0.001\mathrm{M})\right|\left|\mathrm{Y}^{2+}(0.01\mathrm{M})\right|\mathrm{Y}$ is _____ $\times 10^{-2}$ V (Nearest integer)

Given: $\mathrm{E}^{0}_{\mathrm{X}^{2+}|\mathrm{X}} = -2.36\ \mathrm{V}$

$\mathrm{E}^{0}_{\mathrm{Y}^{2+}|\mathrm{Y}} = +0.36\ \mathrm{V}$

$\frac{2.303\mathrm{RT}}{\mathrm{F}} = 0.06\ \mathrm{V}$

Open solution

Q188. JEE Main 2023 (Online) 30th January Morning Shift

Consider the cell

$\mathrm{Pt}_{(\mathrm{s})}\left|\mathrm{H}_2(\mathrm{g}, 1\ \mathrm{atm})\right|\mathrm{H}^{+}(\mathrm{aq}, 1\mathrm{M})\left|\left|\mathrm{Fe}^{3+}(\mathrm{aq}), \mathrm{Fe}^{2+}(\mathrm{aq})\right|\mathrm{Pt}(\mathrm{s})\right.$

When the potential of the cell is 0.712 V at 298 K, the ratio $\left[\mathrm{Fe}^{2+}\right]/\left[\mathrm{Fe}^{3+}\right]$ is _____. (Nearest integer)

Given : $\mathrm{Fe}^{3+} + \mathrm{e}^{-} = \mathrm{Fe}^{2+}, \mathrm{E}^{\theta}\mathrm{Fe}^{3+}, \mathrm{Fe}^{2+}\left|\mathrm{Pt} = 0.771\right.$

$\frac{2.303\mathrm{RT}}{\mathrm{F}} = 0.06\ \mathrm{V}$

Open solution

Q189. JEE Main 2023 (Online) 29th January Evening Shift

The equilibrium constant for the reaction

$\mathrm{Zn(s)} + \mathrm{Sn}^{2+}(\mathrm{aq}) \rightleftharpoons \mathrm{Zn}^{2+}(\mathrm{aq}) + \mathrm{Sn(s)}$ is 1×10^{20} at 298 K. The magnitude of standard electrode potential of $\mathrm{Sn}/\mathrm{Sn}^{2+}$ if $\mathrm{E}^{\Theta}_{\mathrm{Zn}^{2+}/\mathrm{Zn}} = -0.76$ V is _____ $\times 10^{-2}$ V. (Nearest integer)

Given : $\frac{2.303\mathrm{RT}}{\mathrm{F}} = 0.059\ \mathrm{V}$

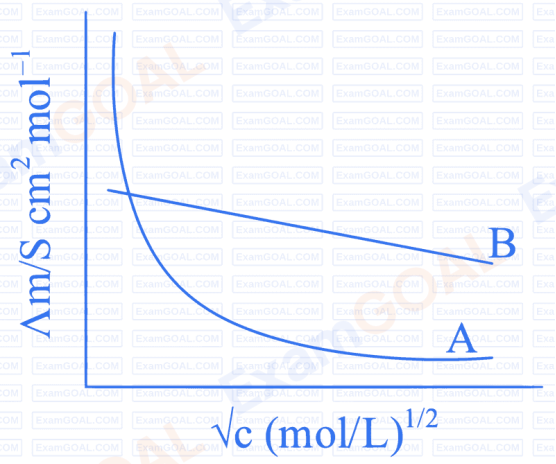
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Q190. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

Following figure shows dependence of molar conductance of two electrolytes on concentration. Λ_m^o is the limiting molar conductivity.



The number of incorrect statement(s) from the following is _____

(A) Λ_m^o for electrolyte A is obtained by extrapolation

(B) For electrolyte B, Λm vs $\sqrt{c}$ graph is a straight line with intercept equal to Λ_m^o

(C) At infinite dilution, the value of degree of dissociation approaches zero for electrolyte B.

(D) Λ_m^o for any electrolyte A and B can be calculated using λ° for individual ions

Q191. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

$Pt(s)|H_2(g)(1\,bar)|H^+(aq)(1\,M)||M^{3+}(aq), M^+(aq)|Pt(s)$

The E_{cell} for the given cell is 0.1115 V at 298 K when $\frac{[M^+(aq)]}{[M^{3+}(aq)]} = 10^a$

The value of a is _____

Given : $E^\theta_{M^{3+}/M^+} = 0.2$ V

$\frac{2.303RT}{F} = 0.059V$

Q192. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

Consider the cell

$Pt(s)|H_2(g)\,(1\,atm)|H^+\,(aq, [H^+] = 1)||Fe^{3+}(aq), Fe^{2+}(aq)|Pt(s)$

Given $E^o_{Fe^{3+}/Fe^{2+}} = 0.771$ V and $E^o_{H^+/1/2\,H_2} = 0\,V$, $T = 298\,K$

If the potential of the cell is 0.712 V, the ratio of concentration of Fe^{2+} to Fe^{3+} is _____
(Nearest integer)

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Q193. JEE Main 2023 (Online) 24th January Morning Shift

At 298 K, a 1 litre solution containing 10 mmol of $Cr_2O_7^{2-}$ and 100 mmol of Cr^{3+} shows a pH of 3.0.

Given : $Cr_2O_7^{2-} \rightarrow Cr^{3+}$; $E^\circ = 1.330V$

and $\frac{2.303\,RT}{F} = 0.059$ V

The potential for the half cell reaction is $x \times 10^{-3}$ V. The value of x is _____

Open solution

Q194. JEE Main 2022 (Online) 30th July Evening Shift

For the given cell $Zn(s)\,|\,Zn^{2+}(C_1\ M)\,||\,Zn^{2+}(C_2\ M)\,|\,Zn(s)$ the change in Gibbs energy (ΔG) will be zero when $\frac{C_1}{C_2}$ is equal to _____.

Open solution

Q195. JEE Main 2022 (Online) 29th July Evening Shift

For a cell, $Cu(s)\,|Cu^{2+}(0.001\ M)||Ag^{+}(0.01\ M)|Ag(s)$

the cell potential is found to be 0.43 V at 298 K. The magnitude of standard electrode potential for Cu^{2+}/Cu is _____ $\times 10^{-2}$ V.

[Given : $E^{\Theta}_{Ag^+/Ag}$ = 0.80 V and $\frac{2.303RT}{F}$ = 0.06 V]

Open solution

Q196. JEE Main 2022 (Online) 29th July Morning Shift

Resistance of a conductivity cell (cell constant $129\ m^{-1}$) filled with 74.5 ppm solution of KCl is $100\ \Omega$ (labelled as solution 1). When the same cell is filled with KCl solution of 149 ppm, the resistance is $50\ \Omega$ (labelled as solution 2). The ratio of molar conductivity of solution 1 and solution 2 is i.e. $\frac{\Lambda_1}{\Lambda_2} = x \times 10^{-3}$. The value of x is _____. (Nearest integer)

Given, molar mass of KCl is $74.5\ g\ mol^{-1}$.

Open solution

Q197. JEE Main 2022 (Online) 26th July Morning Shift

The amount of charge in F (Faraday) required to obtain one mole of iron from Fe_3O_4 is _____. (Nearest Integer)

Open solution

Q198. JEE Main 2022 (Online) 25th July Morning Shift

The cell potential for $Zn\,|Zn^{2+}(aq)|\,|Sn^{x+}|Sn$ is 0.801 V at 298 K. The reaction quotient for the above reaction is 10^{-2}. The number of electrons involved in the given electrochemical cell reaction is _____.

(Given : $E^{o}_{Zn^{2+}|Zn} = -0.763$ V, $E^{o}_{Sn^{x+}|Sn} = +0.008$ V and $\frac{2.303RT}{F} = 0.06$ V)

Open solution

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Q199. JEE Main 2022 (Online) 29th June Evening Shift

The cell potential for the given cell at 298 K

Pt| H_2 (g, 1 bar) | H^+ (aq) || Cu^{2+} (aq) | Cu(s)

is 0.31 V. The pH of the acidic solution is found to be 3, whereas the concentration of Cu^{2+} is 10^{-x} M. The value of x is _____.

(Given : $E^{\Theta}_{Cu^{2+}/Cu}$ = 0.34 V and $\frac{2.303\,RT}{F}$ = 0.06 V)

Open solution

Q200. JEE Main 2022 (Online) 29th June Morning Shift

A dilute solution of sulphuric acid is electrolysed using a current of 0.10 A for 2 hours to produce hydrogen and oxygen gas. The total volume of gases produced a STP is _____ cm^3. (Nearest integer)

[Given : Faraday constant F = 96500 C mol^{-1} at STP, molar volume of an ideal gas is 22.7 L mol^{-1}]

Open solution

Q201. JEE Main 2022 (Online) 28th June Evening Shift

For the given reactions

$Sn^{2+} + 2e^- \rightarrow Sn$

$Sn^{4+} + 4e^- \rightarrow Sn$

the electrode potentials are ; $E^o_{Sn^{2+}/Sn} = -0.140$ V and $E^o_{Sn^{4+}/Sn} = +0.010$ V. The magnitude of standard electrode potential for Sn^{4+}/Sn^{2+} i.e. $E^o_{Sn^{4+}/Sn^{2+}}$ is _____ $\times 10^{-2}$ V. (Nearest integer)

Open solution

Q202. JEE Main 2022 (Online) 28th June Morning Shift

The quantity of electricity in Faraday needed to reduce 1 mol of $Cr_2O_7^{2-}$ to Cr^{3+} is _____.

Open solution

Q203. JEE Main 2022 (Online) 27th June Evening Shift

For the reaction taking place in the cell :

Pt (s)| H_2 (g)|H^+(aq) || Ag^+(aq) |Ag (s)

E^o_{cell} = + 0.5332 V.

The value of $\Delta_f G^\circ$ is _____ kJ mol^{-1}. (in nearest integer)

Open solution

Q204. JEE Main 2022 (Online) 27th June Morning Shift

The limiting molar conductivities of NaI, $NaNO_3$ and $AgNO_3$ are 12.7, 12.0 and 13.3 mS m^2 mol^{-1}, respectively (all at 25°C). The limiting molar conductivity of AgI at this temperature is _____ mS m^2 mol^{-1}.

Open solution

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Q205. JEE Main 2022 (Online) 26th June Evening Shift

Cu(s) + Sn^{2+} (0.001M) $\rightarrow$ Cu^{2+} (0.01M) + Sn(s)

The Gibbs free energy change for the above reaction at 298 K is x $\times 10^{-1}$ kJ mol^{-1}. The value of x is _____. [nearest integer]

[Given : $E^{\Theta}_{Cu^{2+}/Cu} = 0.34\,V$; $E^{\Theta}_{Sn^{2+}/Sn} = -0.14\,V$; F = 96500 C mol^{-1}]

Open solution

Q206. JEE Main 2022 (Online) 25th June Evening Shift

A solution of $Fe_2(SO_4)_3$ is electrolyzed for 'x' min with a current of 1.5 A to deposit 0.3482 g of Fe. The value of x is _____. [nearest integer]

Given : 1 F = 96500 C mol^{-1}

Atomic mass of Fe = 56 g mol^{-1}

Open solution

Q207. JEE Main 2022 (Online) 25th June Morning Shift

In a cell, the following reactions take place

$Fe^{2+} \rightarrow Fe^{3+} + e^-$ $\quad E^o_{Fe^{3+}/Fe^{2+}} = 0.77\,V$

$2I^- \rightarrow I_2 + 2e^-$ $\quad E^o_{I_2/I^-} = 0.54\,V$

The standard electrode potential for the spontaneous reaction in the cell is x $\times 10^{-2}$ V at 298 K. The value of x is _____. (Nearest Integer)

Open solution

Q208. JEE Main 2022 (Online) 24th June Evening Shift

The resistance of a conductivity cell containing 0.01 M KCl solution at 298 K is 1750 Ω. If the conductivity of 0.01 M KCl solution at 298 K is 0.152 $\times 10^{-3}$ S cm^{-1}, then the cell constant of the conductivity cell is _____ $\times 10^{-3}$ cm^{-1}.

Open solution

Q209. JEE Main 2022 (Online) 24th June Morning Shift

The cell potential for the following cell

Pt $|H_2(g)|H^+$ (aq)|| Cu^{2+} (0.01 M)|Cu(s)

is 0.576 V at 298 K. The pH of the solution is _____. (Nearest integer)

(Given : $E^o_{Cu^{2+}/Cu} = 0.34$ V and $\frac{2.303\,RT}{F} = 0.06$ V)

Open solution

Q210. JEE Main 2021 (Online) 1st September Evening Shift

If the conductivity of mercury at 0°C is 1.07 $\times 10^6$ S m^{-1} and the resistance of a cell containing mercury is 0.243Ω, then the cell constant of the cell is x $\times 10^4$ m^{-1}. The value of x is _____. (Nearest integer)

Open solution

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Q211. JEE Main 2021 (Online) 31st August Morning Shift

Consider the following cell reaction :

$$Cd_{(s)} + Hg_2SO_{4(s)} + \frac{9}{5}H_2O_{(l)} \rightleftharpoons CdSO_4.\frac{9}{5}H_2O_{(s)} + 2Hg_{(l)}$$

The value of E^0_{cell} is 4.315 V at 25°C. If $\Delta H^\circ = -825.2$ kJ mol^{-1}, the standard entropy change ΔS° in J K^{-1} is _____. (Nearest integer) [Given : Faraday constant = 96487 C mol^{-1}]

Open solution

Q212. JEE Main 2021 (Online) 27th August Evening Shift

The resistance of a conductivity cell with cell constant 1.14 cm^{-1}, containing 0.001 M KCl at 298 K is 1500 Ω. The molar conductivity of 0.001 M KCl solution at 298 K in S cm^2 mol^{-1} is _____. (Integer answer)

Open solution

Q213. JEE Main 2021 (Online) 26th August Evening Shift

For the galvanic cell,

Zn(s) + Cu^{2+} (0.02 M) $\rightarrow$ Zn^{2+} (0.04 M) + Cu(s),

E_{cell} = _____ $\times 10^{-2}$ V. (Nearest integer)

[Use : $E^0_{Cu/Cu^{2+}}$ = − 0.34 V, $E^0_{Zn/Zn^{2+}}$ = + 0.76 V, $\frac{2.303RT}{F} = 0.059\,V$]

Open solution

Q214. JEE Main 2021 (Online) 26th August Morning Shift

These are physical properties of an element

(A) Sublimation enthalpy

(B) Ionisation enthalpy

(C) Hydration enthalpy

(D) Electron gain enthalpy

The total number of above properties that affect the reduction potential is _____ (Integer answer)

Open solution

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Q215. JEE Main 2021 (Online) 27th July Evening Shift

Open solution

For the cell

Cu(s) | Cu^{2+} (aq) (0.1 M) || Ag^{+}(aq) (0.01 M) | Ag(s)

the cell potential E_1 = 0.3095 V

For the cell

Cu(s) | Cu^{2+} (aq) (0.01 M) || Ag^{+}(aq) (0.001 M) | Ag(s)

the cell potential = _____ $\times 10^{-2}$ V. (Round off to the nearest integer).

[Use : $\frac{2.303RT}{F}$ = 0.059]

Q216. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

The conductivity of a weak acid HA of concentration 0.001 mol L^{-1} is 2.0×10^{-5} S cm^{-1}. If Λ_m^o(HA) = 190 S cm^2 mol^{-1}, the ionization constant (K_a) of HA is equal to _____ $\times 10^{-6}$. (Round off to the Nearest Integer)

Q217. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

Consider the cell at 25°C

Zn | Zn^{2+} (aq), (1M) || Fe^{3+} (aq), Fe^{2+} (aq) | Pt(s)

The fraction of total iron present as Fe^{3+} ion at the cell potential of 1.500 V is x $\times 10^{-2}$. The value of x is _____. (Nearest integer)

(Given : $E^0_{Fe^{3+}/Fe^{2+}} = 0.77V$, $E^0_{Zn^{2+}/Zn} = -0.76V$)

Q218. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

Assume a cell with the following reaction

$$Cu_{(s)} + 2Ag^{+}(1 \times 10^{-3}M) \rightarrow Cu^{2+}(0.250M) + 2Ag_{(s)}$$

$E^{\Theta}_{cell} = 2.97$ V

E_{cell} for the above reaction is _____ V. (Nearest integer)

[Given : log 2.5 = 0.3979, T = 298 K]

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Q219. JEE Main 2021 (Online) 20th July Evening Shift

Open solution

Potassium chlorate is prepared by electrolysis of KCl in basic solution as shown by following equation.

$6OH^- + Cl^- \rightarrow ClO_3^- + 3H_2O + 6e^-$

A current of xA has to be passed for 10h to produce 10.0g of potassium chlorate. The value of x is _____. (Nearest integer)

(Molar mass of $KClO_3$ = 122.6 g mol^{-1}, F = 96500 C)

Q220. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

The molar conductivities at infinite dilution of barium chloride, sulphuric acid and hydrochloric acid are 280, 860 and 426 S cm^2 mol^{-1} respectively. The molar conductivity at infinite dilution of barium sulphate is _____ S cm^2 mol^{-1}. (Round off to the Nearest Integer).

Q221. JEE Main 2021 (Online) 18th March Morning Shift

Open solution

For the reaction

$2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(s)$

the magnitude of the standard molar Gibbs free energy change, $\Delta_r G^o_m$ = – _____ kJ (Round off to the Nearest Integer).

$$\begin{bmatrix} E^o_{Fe^{2+}/Fe(s)} = -0.440V; & E^o_{Fe^{3+}/Fe(s)} = -0.036V \\ E^o_{I_2/2I^-} = 0.539V; & F = 96500C \end{bmatrix}$$

Q222. JEE Main 2021 (Online) 17th March Evening Shift

Open solution

A KCl solution of conductivity 0.14 S m^{-1} shows a resistance of 4.19Ω in a conductivity cell. If the same cell is filled with an HCl solution, the resistance drops to 1.03Ω. The conductivity of the HCl solution is _____ $\times$ 10^{-2} S m^{-1}. (Round off to the Nearest Integer).

Q223. JEE Main 2021 (Online) 16th March Evening Shift

Open solution

A 5.0 m mol dm^{-3} aqueous solution of KCl has a conductance of 0.55 mS when measured in a cell of cell constant 1.3 cm^{-1}. The molar conductivity of this solution is _____ mSm^2 mol^{-1}. (Round off to the Nearest Integer).

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Q224. JEE Main 2021 (Online) 26th February Evening Shift

Open solution

Emf of the following cell at 298K in V is x $\times$ 10^{-2}.

Zn|Zn^{2+}(0.1 M)||Ag^{+} (0.01 M)|Ag

The value of x is _____. (Rounded off to the nearest integer)

[Given : $E^{\theta}_{Zn^{2+}/Zn} = -0.76V; E^{\theta}_{Ag^{+}/Ag} = +0.80V; \frac{2.303RT}{F} = 0.059$]

Q225. JEE Main 2021 (Online) 26th February Morning Shift

Open solution

Consider the following reaction

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{+2} + 4H_2O, E^o = 1.51V.$

The quantity of electricity required in Faraday to reduce five moles of MnO_4^- is _____. (Integer answer)

Q226. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

Copper reduces NO_3^- into NO and NO_2 depending upon the concentration of HNO_3 in solution. (Assuming fixed [Cu^{2+}] and $P_{NO} = P_{NO_2}$), the HNO_3 concentration at which the thermodynamic tendency for reduction of NO_3^- into NO and NO_2 by copper is same is 10^x M. The value of 2x is _____. (Rounded off to the nearest integer)

[Given, $E^o_{Cu^{2+}/Cu} = 0.34$ V, $E^o_{NO_3^-/NO} = 0.96$ V, $E^o_{NO_3^-/NO_2} = 0.79$ V and at 298 K, $\frac{RT}{F}(2.303) = 0.059$]

Q227. JEE Main 2021 (Online) 24th February Evening Shift

Open solution

The magnitude of the change in oxidising power of the MnO_4^-/Mn^{2+} couple is x $\times$ 10^{-4} V, if the H^+ concentration is decreased from 1M to 10^{-4} M at 25°C. (Assume concentration of MnO_4^- and Mn^{2+} to be same on change in H^+ concentration). The value of x is _____.

$\left[Given : \frac{2.303RT}{F} = 0.059\right]$

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Q228. JEE Main 2020 (Online) 6th September Morning Slot

Open solution

Potassium chlorate is prepared by the electrolysis of KCl in basic solution

$6OH^- + Cl^- \rightarrow ClO_3^- + 3H_2O + 6e^-$

If only 60% of the current is utilized in the reaction, the time (rounded to the nearest hour) required to produce 10 g of $KClO_3$ using a current of 2 A is_____.

(Given : F = 96,500 C mol^{-1}; molar mass of $KClO_3$ = 122 g mol^{-1})

Q229. JEE Main 2020 (Online) 5th September Morning Slot

Open solution

An oxidation-reduction reaction in which
3 electrons are transferred has a ΔG^o of 17.37 kJ mol^{-1} at
25 ^oC. The value of E^o
cell (in V) is _____ × 10^{-2}.
(1 F = 96,500 C mol^{-1})

Q230. JEE Main 2020 (Online) 3rd September Evening Slot

Open solution

An acidic solution of dichromate is electrolyzed for 8 minutes using 2A current. As per the following equation

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

The amount of Cr^{3+} obtained was 0.104 g. The efficiency of the process(in%) is (Take : F = 96000 C, At. mass of chromium = 52) _____.

Q231. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

The photoelectric current from Na (Work function, w_0 = 2.3 eV) is stopped by the output voltage of the cell
Pt(s) | H_2 (g, 1 Bar) | HCl (aq., pH =1) | AgCl(s) | Ag(s).
The pH of aq. HCl required to stop the photoelectric current from K(w_0 = 2.25 eV), all other conditions remaining the same, is _____ × 10^{-2} (to the nearest integer).

Given, $2.303\frac{RT}{F}$ = 0.06 V;
$E^0_{AgCl|Ag|Cl^-}$ = 0.22 V

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Q232. JEE Main 2020 (Online) 2nd September Evening Slot

Open solution

For the disproportionation reaction
$2Cu^{+}(aq) \rightleftharpoons Cu(s) + Cu^{2+}(aq)$ at 298 K. ln K (where K is the equilibrium constant) is _____ $\times 10^{-1}$.
Given :
($E^{0}_{Cu^{2+}/Cu^{+}} = 0.16V$
$E^{0}_{Cu^{+}/Cu} = 0.52V$
$\frac{RT}{F} = 0.025$)

Q233. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution

The Gibbs change (in J) for the given reaction at $[Cu^{2+}] = [Sn^{2+}] = 1$ M and 298K is :

$Cu(s) + Sn^{2+}(aq.) \rightarrow Cu^{2+}(aq.) + Sn(s)$;

($E^{0}_{Sn^{2+}|Sn} = -0.16\,V$,
$E^{0}_{Cu^{2+}|Cu} = 0.34\,V$)
Take F = 96500 C mol^{-1})

Q234. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

108 g of silver (molar mass 108 g mol^{-1}) is deposited at cathode from $AgNO_3(aq)$ solution by a certain quantity of electricity. The volume (in L) of oxygen gas produced at 273 K and 1 bar pressure from water by the same quantity of electricity is _____.

Q235. JEE Main 2020 (Online) 8th January Evening Slot

Open solution

For an electrochemical cell

$Sn(s) \mid Sn^{2+}(aq,1M) \| Pb^{2+}(aq,1M) \mid Pb(s)$

the ratio $\frac{[Sn^{2+}]}{[Pb^{2+}]}$ when this cell attains equilibrium is _____.

(Given $E^{0}_{Sn^{2+}|Sn} = -0.14V$,

$E^{0}_{Pb^{2+}|Pb} = -0.13V$, $\frac{2.303RT}{F} = 0.06$)

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Q236. JEE Main 2020 (Online) 8th January Morning Slot

Open solution

What would be the electrode potential for the given half cell reaction at pH = 5? _____.

$2H_2O \rightarrow O_2 + 4H^{\oplus} + 4e^-$; E^0_{red} = 1.23 V

(R = 8.314 J mol^{-1} K^{-1} ; Temp = 298 K;

oxygen under std. atm. pressure of 1 bar)

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **D**	3. **A**	4. **B**	5. **A**
6. **B**	7. **A**	8. **D**	9. **D**	10. **C**
11. **D**	12. **C**	13. **B**	14. **B**	15. **A**
16. **A**	17. **D**	18. **B**	19. **C**	20. **C**
21. **A**	22. **A**	23. **D**	24. **C**	25. **A**
26. **C**	27. **C**	28. **D**	29. **B**	30. **B**
31. **D**	32. **A**	33. **D**	34. **D**	35. **D**
36. **D**	37. **B**	38. **A**	39. **D**	40. **C**
41. **B**	42. **C**	43. **D**	44. **C**	45. **D**
46. **B**	47. **C**	48. **D**	49. **C**	50. **C**
51. **C**	52. **C**	53. **B**	54. **A**	55. **B**
56. **C**	57. **A**	58. **A**	59. **D**	60. **B**
61. **A**	62. **B**	63. **C**	64. **D**	65. **D**
66. **B**	67. **D**	68. **D**	69. **A**	70. **D**
71. **D**	72. **C**	73. **A**	74. **D**	75. **B**
76. **A**	77. **C**	78. **A**	79. **D**	80. **B**
81. **D**	82. **C**	83. **A**	84. **D**	85. **A**
86. **B**	87. **A**	88. **C**	89. **D**	90. **D**
91. **D**	92. **D**	93. **C**	94. **C**	95. **A**
96. **A**	97. **C**	98. **B**	99. **B**	100. **C**
101. **C**	102. **C**	103. **C**	104. **C**	105. **D**
106. **B**	107. **A**	108. **C**	109. **C**	110. **B**
111. **C**	112. **D**	113. **D**	114. **C**	115. **C**
116. **C**	117. **C**	118. **A**	119. **B**	120. **C**
121. **D**	122. **D**	123. **D**	124. **A**	125. **D**
126. **A**	127. **C**	128. **D**	129. **C**	130. **C**
131. **A**	132. **C**	133. **C**	134. **C**	135. **D**
136. **A**	137. **B**	138. **B**	139. **B**	140. **C**
141. **B**	142. **A**	143. **D**	144. **C**	

Numerical

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145. **15**	146. **2**	147. **560**	148. **78**	149. **4**
150. **24**	151. **2**	152. **111**	153. **4**	154. **4**
155. **200**	156. **6**	157. **10**	158. **400**	159. **3**
160. **23**	161. **6**	162. **2**	163. **150**	164. **3**
165. **2**	166. **338**	167. **1**	168. **4**	169. **5**
170. **2**	171. **46**	172. **18**	173. **108**	174. **3**
175. **25**	176. **161**	177. **4**	178. **2**	179. **66**
180. **1825**	181. **1**	182. **3**	183. **13039**	184. **3**
185. **25**	186. **6**	187. **275**	188. **10**	189. **17**
190. **2**	191. **3**	192. **10**	193. **917**	194. **1**
195. **34**	196. **1000**	197. **3**	198. **4**	199. **7**
200. **127**	201. **16**	202. **6**	203. **51 or 103**	204. **14**
205. **983**	206. **20**	207. **23**	208. **266**	209. **5**
210. **26**	211. **25**	212. **760**	213. **109**	214. **3**
215. **28**	216. **12**	217. **24**	218. **3**	219. **1**
220. **288**	221. **46**	222. **57**	223. **14**	224. **147**
225. **25**	226. **1**	227. **3776**	228. **11**	229. **-6**
230. **60**	231. **142**	232. **144**	233. **96500**	234. **5.66 to 5.68**
235. **2.13 to 2.16**	236. **1.52 to 1.53**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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